Conformal Risk Control for Foundation Models Under Prompt Shift: What Weighting Adds to the Online Update

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Abstract

Prompts sent to a foundation model change over time, and a risk threshold calibrated on old prompts can stop being valid. We study PromptShift-CRC, a rule that weights past prompts by feature similarity, falls back to recent prompts when the current prompt is unlike the past, and updates its level online. We prove that the long-run violation rate is controlled by the update for any weights, so marginal coverage cannot separate weighting schemes. We give a conditional bound at a fixed level whose cost is a weighted total variation distance, bounded by a computable prompt-drift term up to an unknown Lipschitz constant. We propose a protocol for comparing such rules: a matched short-window baseline, a validity and sharpness frontier, 30-seed synthetic streams, and real-output streams ordered without using the model's output. On MMLU and TruthfulQA (Qwen2.5-1.5B) and RealToxicityPrompts (GPT-2, replicated with a second generator and classifier), every online method reaches the target. PromptShift-CRC stays within 0.0055 of the Adaptive CRC frontier, avoids the worst window, and is matched or beaten by DtACI in most settings. Its practical value is that it performs like a mid-ranked window without having to choose one. Tests fixed in advance to give the weights a better chance (sharper kernels, regimes that recur, a second moving target, a published localised competitor) show no consistent gain over the frontier. Code, outputs and streams are released.

Keywords: conformal risk control; prompt shift; foundation models; online conformal prediction; uncertainty quantification.

1 Introduction

A foundation model in use sees a stream of prompts that changes over time. New users arrive, topics change, policies change, and the model meets kinds of request that were rare when its risk was first measured. Conformal prediction and conformal risk control give a way to set a threshold on a risk score so that the long-run rate of violations stays near a chosen level α, with no assumption on the model (Vovk et al., 2005; Angelopoulos and Bates, 2021; Angelopoulos et al., 2024b). The guarantee is clearest when the calibration examples and the future examples are exchangeable, and a prompt stream with drift is the case where they are not.

Two families of tools address this. Weighted conformal methods change the calibration distribution with importance weights (Tibshirani et al., 2019; Barber et al., 2023). Online methods

adjust the target level after each observed violation (Gibbs and Candès, 2021, 2024; Angelopoulos et al., 2023). A natural proposal for foundation models is to combine them: weight past prompts by how similar they are to the current prompt, fall back to recent prompts when the current prompt looks unlike the past, and update the level online. We call this combination PromptShift-CRC. This paper asks what each part of such a method contributes, and it answers with theory, a fixed evaluation protocol, and experiments on real model outputs.

What we find. The answer is more modest than we expected.

(i) With the online update used here, long-run risk control is a property of the update and not of the weights. For every weighting scheme the average violation rate is within $(\max\{\alpha, 1-\alpha\}+\gamma)/(T\gamma)$ of α (Theorem 2), so marginal coverage cannot tell weighting schemes apart. In all of our experiments, every method that uses this update is within the bound, as Theorem 2 requires; DtACI and quantile tracking, which use other updates, are within it as well.

(ii) The weights do matter for *local* behaviour and for sharpness. We give a bound at a fixed level in which the cost of using calibration points from a different regime is a weighted sum of total variation distances (Theorem 1) and, under a smoothness condition, a computable *prompt drift* D_t built from the weights actually used (Corollary 1). This is a statement about how a penalty scales and not a certificate.

(iii) A fair comparison needs a baseline that uses a short calibration window. We call Adaptive conformal inference with a window equal to PromptShift-CRC's recent window the *matched* baseline, and we also trace the validity and sharpness trade-off of Adaptive CRC by sweeping its window. In our experiments PromptShift-CRC and its variants stay within a small margin of Adaptive CRC: on average over streams, the post-shift violation of PromptShift-CRC differs from the Adaptive CRC curve at equal sharpness by at most 0.0055 in every setting. DtACI does as well or better in most. The one advantage of PromptShift-CRC is that it performs like a mid-ranked window without a window having to be chosen (Section 6.3). Tests fixed in advance to give the weights a better chance do not change this (Sections 6.2 to 6.7).

(iv) These results hold on synthetic streams with 30 seeds each and on streams built from real model outputs (MMLU and TruthfulQA with a 1.5-billion-parameter model, and RealToxicityPrompts with GPT-2 continuations scored by a public toxicity classifier, replicated with a second generator and classifier), ordered by variables that do not depend on the model's output.

Contributions.

(1) A statement of the PromptShift-CRC rule, in which the weights, the drift gate and the recency fallback depend only on prompt features, with the finite-sample correction needed for a valid conditional bound (Section 3).

(2) A conditional guarantee at a fixed level, a drift penalty computed from the final weights, a long-run guarantee for the online rule that holds for any weights, and a bound on its conditional violation probability at the running level, with proofs (Section 4).

(3) An evaluation protocol with a matched-window baseline and a window sweep, a validity and sharpness frontier, multi-seed synthetic streams, orderings that do not use the outcome, real model outputs and, for the real streams, settings fixed before the results were seen (Section 5).

(4) An empirical answer, including where the drift-aware components help a little, where they do not, and what a practitioner should use (Section 6).

2 Related work

Conformal prediction and conformal risk control. Split conformal prediction gives finite-sample coverage under exchangeability (Vovk et al., 2005; Angelopoulos and Bates, 2021). Conformal risk control extends it from coverage to any bounded monotone loss (Angelopoulos et al., 2024b). Under covariate shift, weighting the calibration scores by the likelihood ratio restores coverage when the ratio is known (Tibshirani et al., 2019); Barber et al. (2023) show that fixed weights lose coverage by an amount governed by the total variation distance between the data distribution and a swapped version of it, which is the form of guarantee we use. Angelopoulos et al. (2024b) also give a weighted version of conformal risk control under covariate shift, and Gibbs et al. (2023) obtain coverage conditional on features over a class of covariate shifts. Cauchois et al. (2024) and Podkopaev and Ramdas (2022) treat other kinds of shift.

Online conformal methods. Adaptive conformal inference updates the miscoverage level after each step and has a deterministic long-run bound for any data sequence (Gibbs and Candès, 2021). Later work chooses the step size automatically or makes it decay, aggregates several step sizes, tracks the quantile directly, or adds integral and forecasting terms (Zaffran et al., 2022; Gibbs and Candès, 2024; Bhatnagar et al., 2023; Angelopoulos et al., 2024a, 2023). For risk control with losses beyond coverage, Feldman et al. (2023) give an online procedure, Zecchin and Simeone (2024) make the control local by letting the threshold depend on features, and Timans et al. (2025) monitor risk violations under unknown shift. Our long-run result is the Gibbs and Candès bound applied to our update, and we do not claim it as new.

Weighting and localisation under shift. Similarity weights, localised calibration and density-ratio weights are standard ways to adapt a calibration set to the current input (Guan, 2023; Bastani et al., 2022; Zecchin et al., 2025). For language models, Mohri and Hashimoto (2024) give conformal factuality guarantees, Lin et al. (2026) weight calibration scores for domain shift, and Hu et al. (2026) estimate density ratios online to control hallucination under covariate shift. PromptShift-CRC is closest to these: it is a feature-based weighting rule with a recency fallback and an online level update. We do not claim a new weighting idea. Our contribution is an account of what such a rule can and cannot guarantee, and a protocol that compares it with strong, simple baselines.

Language-model uncertainty and benchmarks. Conformal ideas have been applied to multiple-choice question answering, generation and open-ended outputs (Kumar et al., 2023; Wang et al., 2025; Noorani et al., 2025). We use MMLU and TruthfulQA (Lin et al., 2021). Earlier versions of this work also used streams built from RealToxicityPrompts, SummEval and LongHalQA (Gehman et al., 2020; Fabbri et al., 2021; Qiu et al., 2024). Those streams contain no model outputs and are reported in the appendix with that label. Evidence that prompts and model behaviour shift in practice comes from studies of code tasks, semantic rewording and non-deterministic drift (Li et al., 2024, 2025; Nicholson, 2026).

3 Setting and method

At step t the system receives a prompt with feature vector $E_t \in \mathbb{R}^d$ (a sentence embedding of the prompt), a fixed model produces a response, and a risk score $R_t \in [0, 1]$ is revealed after the decision. Large values mean more risk, for example $R_t = 1 - p_t$ where p_t is the probability the model gave to the correct answer, or the 0/1 error. The aim is a threshold q_t, chosen from $E_{1:t}$ and $R_{1:t-1}$, so that the violation indicator $\mathrm{err}_t = \mathbf{1}\{R_t > q_t\}$ has long-run mean close to α and, more usefully, has no long stretch with a much higher rate. With the indicator loss this is the conformal-prediction case of conformal risk control (Angelopoulos et al., 2024b) and our guarantees and experiments cover it. A threshold above a cut-off τ triggers an action such as abstaining, retrieving or asking for review. We write the fraction of such steps as the *intervention rate*.

Weights. Given a pool of past steps $s \in \mathcal{C}_t = \{t - 3W, \ldots, t - 1\}$ (with W a window length), the method forms kernel weights from the distance of each past prompt to the current prompt,

$$\tilde{s}_{t,s} = \exp\{-\|E_s - E_t\|^2/h^2\} + \varepsilon, \qquad s_{t,s} = \tilde{s}_{t,s} \Big/ \textstyle\sum_{j \in \mathcal{C}_t} \tilde{s}_{t,j},$$

with bandwidth $h > 0$ and a small floor ε for numerical safety. The *pre-gate drift* is the weighted mean distance $D_t^{\mathrm{pre}} = \sum_s s_{t,s}\|E_s - E_t\|$. A logistic gate $g_t = \{1 + \exp[\kappa(D_t^{\mathrm{pre}} - c)]\}^{-1} \in (0, 1)$ mixes the kernel weights with uniform weights on the most recent R steps $\mathcal{R}_t \subseteq \mathcal{C}_t$:

$$p_{t,s} \;\propto\; g_t\, s_{t,s} + (1 - g_t)\, \frac{\mathbf{1}\{s \in \mathcal{R}_t\}}{|\mathcal{R}_t|} + \varepsilon, \qquad \textstyle\sum_s p_{t,s} = 1.$$

When the current prompt is close to many past prompts (D_t^{pre} small) the gate is near one and the rule borrows from all similar prompts. When it is far from the past, the rule leans on the most recent steps. The weights depend on prompt features only and never on scores. We report the *post-gate drift* $D_t = \sum_s p_{t,s}\|E_s - E_t\|$ and the effective size $n_{\mathrm{eff},t} = 1/\sum_s p_{t,s}^2$.

Threshold and update. The threshold is the quantile of the weighted past scores at level $\ell_t = (1 - \alpha_t)(1 + m_t)$, with $m_t = \max_s p_{t,s}$ and the conventions of Section 4, and $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathrm{err}_t)$ with $\alpha_1 = \alpha$. The update has no projection. The factor $1 + m_t$ is the Barber et al. correction and equals the usual $(1 + 1/n)$ for uniform weights.

Baselines and variants. *Static*: the first calibration segment only. *Rolling*: uniform weights on the last W steps, fixed level. *Adaptive CRC*: rolling window plus the online update, run with the full window W and with a window equal to the recent window R (*matched*), and with a sweep of other windows. *Representation-weighted*: kernel weights over a pool of $3W$ steps, with (`rep_aci`) and without the update. *PromptShift-CRC*: the full rule above. *Ablations*: no kernel weights (uniform weights inside the gate), no gate (fixed mix $g_t = 0.5$), no recency fallback ($g_t \equiv 1$), and no online update. Three published online methods serve as stronger baselines: quantile tracking (the P term of conformal PID control, Angelopoulos et al., 2023), DtACI (Gibbs and Candès, 2024) with its recommended constants, and a density-ratio weighted rule in the spirit of Lin et al. (2026); Hu et al. (2026) (simplified: a logistic regression separating the recent window from the older pool, refit every 25 steps). Localised adaptive risk control (Zecchin and Simeone, 2024) is compared in Section 6.6, with the authors' update and defaults.

4 Theory

We use the notation of Section 3: weights $p_{t,s}$ that depend on the features $E_{1:t}$ only, $m_t = \max_{s<t} p_{t,s}$ and $n_{\text{eff},t} = 1/\sum_s p_{t,s}^2$.

The threshold. For a miscoverage level α_t the threshold is the weighted quantile of the past scores at level

$$\ell_t = (1-\alpha_t)(1+m_t), \qquad q_t = \inf\Big\{x : \ \textstyle\sum_{s<t} p_{t,s}\mathbf{1}\{R_s \le x\} \ge \ell_t\Big\},$$

with the conventions $q_t = +\infty$ if $\ell_t > 1$ and $q_t = -\infty$ if $\ell_t \le 0$. The factor $1+m_t$ is the finite-sample correction of Barber et al. (2023): it equals the quantile, at level $1-\alpha_t$, of $\sum_{s<t} \tilde{w}_{t,s}\delta_{R_s} + \tilde{w}_{t,t}\delta_{+\infty}$ with $\tilde{w}_{t,s} = p_{t,s}/(1+m_t)$ and $\tilde{w}_{t,t} = m_t/(1+m_t)$. For uniform weights on n scores this is the usual split-conformal level $(1-\alpha)(1+1/n)$. The level follows the online rule $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathrm{err}_t)$, $\mathrm{err}_t = \mathbf{1}\{R_t > q_t\}$, $\alpha_1 = \alpha$, *with no projection.*

4.1 A conditional guarantee at a fixed level

Assumption 1 (Conditional independence)**.** *Given the features $E_{1:t}$, the scores $R_1, \dots, R_t$ are independent and $R_s \mid E_{1:t} \sim \kappa(E_s, \cdot)$ for a fixed Markov kernel κ (the law of the risk of a prompt with feature e). Write $L_s = \kappa(E_s, \cdot)$.*

Assumption 1 holds, for example, when prompts are independent draws and the model's randomness is independent across prompts. It does not require the laws L_s to be equal.

Theorem 1 (Conditional coverage at a fixed level)**.** *Fix t and $\alpha \in (0,1)$ and let q_t^+ be the threshold above with $\alpha_t \equiv \alpha$. Under Assumption 1,*

$$\mathbb{P}\big(R_t > q_t^+ \mid E_{1:t}\big) \ \le \ \alpha + \frac{2}{1+m_t}\sum_{s<t} p_{t,s}\, d_{\mathrm{TV}}(L_s, L_t) \ \le \ \alpha + 2\sum_{s<t} p_{t,s}\, d_{\mathrm{TV}}(L_s, L_t).$$

In particular, if $L_s = L_t$ for every s with $p_{t,s} > 0$, then $\mathbb{P}(R_t > q_t^+ \mid E_{1:t}) \le \alpha$.

Proof. Condition on $E_{1:t} = e$; the weights $p_{t,s}$ are then fixed numbers. Apply Theorem 2 of Barber, Candès, Ramdas and Tibshirani (2023) and the remark following it (their Section 4.1) that the same bound holds for nonexchangeable split conformal prediction, with the scores R_s in place of the residuals and no fitted function. The sequence is $(R_1, \dots, R_t)$ with conditional joint law $\mathcal{L}(R_{1:t} \mid e)$; the fixed weights are $\hat{w}_s = p_{t,s}/m_t \in [0,1]$ for $s < t$ (their Section 3.1) and the test weight is 1. Their normalised weights (their equation (10)) are $\hat{w}_s/(\sum_{j<t}\hat{w}_j + 1) = p_{t,s}/(1+m_t) = \tilde{w}_{t,s}$, and their threshold (their equation (11)) is the quantile q_t^+ defined above. The theorem gives $\mathbb{P}(R_t \le q_t^+ \mid e) \ge 1 - \alpha - \sum_{s<t}\tilde{w}_{t,s}\, d_{\mathrm{TV}}(\mathbf{R}, \mathbf{R}^s)$, where $\mathbf{R} = (R_1, \dots, R_t)$ and $\mathbf{R}^s$ is $\mathbf{R}$ with entries s and t swapped. Under Assumption 1, $\mathbf{R}$ has product law $\bigotimes_j L_j$ and $\mathbf{R}^s$ has the product law with L_s and L_t exchanged, and these differ only in coordinates s and t, so $d_{\mathrm{TV}}(\mathbf{R}, \mathbf{R}^s) \le d_{\mathrm{TV}}(L_s \otimes L_t,\ L_t \otimes L_s) \le 2\, d_{\mathrm{TV}}(L_s, L_t)$ (a product of two laws changes by at most the sum of the coordinate changes, by the triangle inequality). Substituting gives the first bound; the second uses $1/(1+m_t) \le 1$. □

Theorem 1 holds at a fixed level and does not apply to the data-dependent level α_t of the online rule; Theorem 2 below covers that rule for every choice of weights, including weights that depend on past scores, and, for weights that depend on the features only, Proposition 1 bounds its conditional violation probability at the running level.

Lemma 1 (Size of the correction)**.** $1/n_{\mathrm{eff},t} \le m_t \le n_{\mathrm{eff},t}^{-1/2}$.

Proof. $\sum_s p_{t,s}^2 \le m_t \sum_s p_{t,s} = m_t$ gives the first inequality, and $m_t^2 \le \sum_s p_{t,s}^2$ gives the second. □

The correction factor in ℓ_t therefore lies between $1+1/n_{\mathrm{eff},t}$ and $1+n_{\mathrm{eff},t}^{-1/2}$: concentrated weights are charged in the level, not only in the variance of the weighted quantile.

Corollary 1 (Drift controls the penalty)**.** *Suppose, in addition,* $d_{\mathrm{TV}}\big(\mathcal{L}(R \mid E = e), \mathcal{L}(R \mid E = e')\big) \le \Lambda\|e - e'\|$ *for all* e, e' *in the support of the features. Then, for the final (post-gate) weights,*

$$\mathbb{P}(R_t > q_t^+ \mid E_{1:t}) \le \alpha + \frac{2\Lambda}{1+m_t} D_t, \qquad D_t = \sum_{s<t} p_{t,s} \, \|E_s - E_t\|.$$

The quantity D_t is the *prompt drift* of the paper, computed with the weights that are actually used to form the threshold (after the gate and recency mixing), so the diagnostic is the one that appears in the bound. The Lipschitz constant Λ is a property of the data-generating process; it is not estimated by the method, and the bound is a statement about how the penalty scales, not a computable certificate.

Corollary 1 and Lemma 1 show the trade-off that any drift-aware rule faces: concentrating weight on prompts close to E_t lowers D_t and raises m_t (lowers $n_{\mathrm{eff},t}$). Appendix A gives the sampling error of the weighted quantile in the no-drift case, which grows as $n_{\mathrm{eff},t}^{-1/2}$. Neither bound orders the rules, so the comparison between rules is empirical (Section 6).

4.2 A long-run guarantee for the online update

Theorem 2 (Long-run control, any weights)**.** *For every* $T \ge 1$*, every score sequence* $R_1, \dots, R_T \in [0,1]$ *and every choice of weights,*

$$\Big|\frac{1}{T}\sum_{t=1}^{T} \mathrm{err}_t - \alpha\Big| \;\le\; \frac{\max\{\alpha_1, 1-\alpha_1\} + \gamma}{T\gamma}.$$

Proof. We first show $\alpha_t \in [-\gamma, 1+\gamma]$ for all t, by induction from $\alpha_1 = \alpha \in (0,1)$. Suppose $\alpha_t \in [-\gamma, 1+\gamma]$. If $\alpha_t < 0$ then $\ell_t = (1-\alpha_t)(1+m_t) > 1$, so $q_t = +\infty$, $\mathrm{err}_t = 0$ and $\alpha_{t+1} = \alpha_t + \gamma\alpha \in (\alpha_t, \gamma\alpha) \subset [-\gamma, 1+\gamma]$. If $\alpha_t > 1$ then $\ell_t < 0$, so $q_t = -\infty$, $\mathrm{err}_t = 1$ and $\alpha_{t+1} = \alpha_t - \gamma(1-\alpha) \in [1-\gamma(1-\alpha), \alpha_t) \subset [-\gamma, 1+\gamma]$. If $\alpha_t \in [0,1]$ then $\alpha_{t+1} \in [-\gamma(1-\alpha),\, 1+\gamma\alpha] \subset [-\gamma, 1+\gamma]$. Summing the update, $\alpha_{T+1} - \alpha_1 = \gamma \sum_{t \le T}(\alpha - \mathrm{err}_t)$, so $\frac{1}{T}\sum_t \mathrm{err}_t - \alpha = (\alpha_1 - \alpha_{T+1})/(T\gamma)$, and $\alpha_1 - \alpha_{T+1} \le \alpha_1 + \gamma$ and $\alpha_{T+1} - \alpha_1 \le 1 + \gamma - \alpha_1$, so $|\alpha_1 - \alpha_{T+1}| \le \max\{\alpha_1, 1-\alpha_1\} + \gamma$, which gives the bound. □

This is Proposition 4.1 of Gibbs and Candès (2021) specialised to our update; we give the proof because the two conventions at the extremes ($\ell_t > 1$ and $\ell_t < 0$) are what the proof uses, and the correction factor $1+m_t$ leaves them unchanged. The bound is the same for every weighting scheme: *long-run marginal coverage is a property of the update, not of PromptShift-CRC's weights.* The bound is 0.018 to 0.024 for the synthetic streams, 0.020 for MMLU, 0.012 for RealToxicityPrompts and 0.043 for TruthfulQA.

Proposition 1 (Conditional bound at the running level)**.** *Suppose that every law* L_s *with* $s < t$ *and* $p_{t,s} > 0$ *is continuous. Fix* t*,* $\eta \in (0,1)$ *and an integer* $M \ge 1$*, and write*

$$\varepsilon_t = \sqrt{\frac{\log(2M/\eta)}{2\, n_{\mathrm{eff},t}}} + \frac{1}{M}.$$

Under Assumption 1, conditionally on $E_{1:t}$, with probability at least $1-\eta$ over $R_{1:t-1}$,

$$\mathbb{P}\big(R_t > q_t \;\big|\; E_{1:t}, R_{1:t-1}\big) \;\le\; \max\{\alpha_t, 0\} + \sum_{s<t} p_{t,s}\, d_{\mathrm{TV}}(L_s, L_t) + \varepsilon_t,$$

where q_t is the threshold of the online rule at its own level α_t.

Proof. Condition on $E_{1:t}$, so the weights $p_{t,s}$ are fixed numbers. The level α_t and the threshold q_t are functions of $R_{1:t-1}$ (and of the features), and under Assumption 1 the score R_t is independent of $R_{1:t-1}$ with law L_t. Hence $\mathbb{P}(R_t > q_t \mid E_{1:t}, R_{1:t-1}) = 1 - F_t(q_t)$, where F_s is the distribution function of L_s.

Let $\bar{F} = \sum_{s<t} p_{t,s} F_s$, which is continuous because every F_s in the sum is, and $\hat{F}(x) = \sum_{s<t} p_{t,s} \mathbf{1}\{R_s \le x\}$. For a fixed x, $\hat{F}(x) - \bar{F}(x)$ is a sum of independent, mean-zero terms $p_{t,s}(\mathbf{1}\{R_s \le x\} - F_s(x))$, each in an interval of length $p_{t,s}$, so Hoeffding's inequality gives $\mathbb{P}(|\hat{F}(x) - \bar{F}(x)| > \varepsilon) \le 2\exp(-2\varepsilon^2 n_{\mathrm{eff},t})$. The grid argument in the proof of Proposition 2 uses only independence of the summands, continuity of the target function (so that $\bar{F}(x_j) = j/M$) and the absence of atoms in each L_s (so that almost surely no R_s equals any x_j). The grid $x_j = \bar{F}^{-1}(j/M)$ depends on $E_{1:t}$ only, so it gives an event $\mathcal{E}$ with $\mathbb{P}(\mathcal{E} \mid E_{1:t}) \ge 1 - \eta$ on which $\sup_x |\hat{F}(x) - \bar{F}(x)| \le \varepsilon_t$. The event $\mathcal{E}$ does not involve α_t or q_t, and the supremum is over all x, so the bound holds at the random point q_t.

On $\mathcal{E}$ there are three cases. If $\ell_t > 1$ then $q_t = +\infty$, $1 - F_t(q_t) = 0$, and the right-hand side is nonnegative; this case contains every $\alpha_t < 0$, which the unprojected update can produce (Theorem 2), and without the positive part the bound could fail there. If $\ell_t \le 0$ then $\alpha_t \ge 1$ and the claim is trivial because $1 - F_t(q_t) \le 1 \le \alpha_t$. If $0 < \ell_t \le 1$ then q_t is finite, $\hat{F}(q_t) \ge \ell_t$ by the definition of the quantile and right-continuity of $\hat{F}$, and $\alpha_t \in [0, 1)$ (since $\alpha_t < 0$ would force $\ell_t > 1$) gives $\ell_t = (1 - \alpha_t)(1 + m_t) \ge 1 - \alpha_t = 1 - \max\{\alpha_t, 0\}$. Then

$$F_t(q_t) = \bar{F}(q_t) + \sum_{s<t} p_{t,s}\big(F_t(q_t) - F_s(q_t)\big) \;\ge\; \hat{F}(q_t) - \varepsilon_t - \sum_{s<t} p_{t,s}\, d_{\mathrm{TV}}(L_s, L_t) \;\ge\; 1 - \alpha_t - \varepsilon_t - \sum_{s<t} p_{t,s}\, d_{\mathrm{TV}}(L_s, L_t),$$

because $|F_t(x) - F_s(x)| \le d_{\mathrm{TV}}(L_s, L_t)$ for every x. Rearranging gives the claim. □

Taking expectations over $R_{1:t-1}$, $\mathbb{P}(R_t > q_t \mid E_{1:t}) \le \mathbb{E}[\max\{\alpha_t, 0\} \mid E_{1:t}] + \sum_{s<t} p_{t,s} d_{\mathrm{TV}}(L_s, L_t) + \varepsilon_t + \eta$. Against Theorem 1 the drift term enters with coefficient 1 instead of $2/(1 + m_t)$, and $\varepsilon_t + \eta$ is the price of the data-dependent level. The proof uses only that α_t is a function of the past, so the bound holds for any history-dependent level and any feature-measurable weights. Keeping the factor $1 + m_t$ would improve the finite-threshold case by $m_t(1 - \alpha_t) \le n_{\mathrm{eff},t}^{-1/2}$ (Lemma 1), which we omit. The bound does not apply to the 0/1 error risk, whose distribution has atoms.

Remark 1 (What the bound does and does not say)**.** *The bound is relative to the running level α_t, not to α, and it is a structural statement and not a certificate: for $n_{\mathrm{eff},t} = 375$, $M = 100$ and $\eta = 0.05$ the sampling term is $\varepsilon_t \approx 0.115$ ($n_{\mathrm{eff},t} = 375$ is the mean effective size of PromptShift-CRC on MMLU). The drift term can fall (under the Lipschitz condition of Corollary 1) and the sampling term rises (Lemma 1) when weight is concentrated on prompts close to E_t. The bound does not say that α_t stays close to α. For that we have Theorem 2 and the following identity, which follows by summing the update over the window, as in the proof of Theorem 2: for any window $a \le t \le b$ of length $L = b - a + 1$,*

$$\frac{1}{L}\sum_{t=a}^{b} \mathrm{err}_t - \alpha = \frac{\alpha_a - \alpha_{b+1}}{L\gamma}.$$

The local violation rate of the online rule over a window is therefore α plus the drop $\alpha_a - \alpha_{b+1}$ of the level over that window, divided by $L\gamma$. For example, at $\alpha = 0.1$ and $\gamma = 0.02$ a maximum local violation of 0.20 over $L = 120$ steps is a drop of 0.24.

Remark 2 (What gating can and cannot do)**.** *Theorem 2 gives every weighting scheme the same long-run bound, so long-run marginal coverage cannot distinguish schemes. By Theorem 1 and Lemma 1, a gate that moves weight toward recent or similar prompts can reduce the drift penalty D_t while it reduces $n_{\mathrm{eff},t}$. We therefore do not claim that the gate or the representation weights improve risk control; whether they improve local validity or sharpness is measured, not assumed.*

5 Evaluation protocol

An earlier version of this work compared PromptShift-CRC with three baselines on single runs of hand-built streams. On re-examination we found that a short-window baseline matches it, that single runs hide the seed-to-seed spread, and that the benchmark streams used no model outputs. This section describes what we do instead. Every real-stream setting was fixed before its results were computed; the synthetic constants are inherited from the earlier version and were chosen while those generators were built (Section 8).

Metrics. Marginal coverage ($1-$ mean of err_t) is reported, but Theorem 2 bounds its distance from $1-\alpha$ for every online method, so it cannot rank them. The metrics that can are the following.

- *Post-shift violation*: the mean of err_t over the first L steps after each known regime boundary ($L = 150$ for MMLU, RealToxicityPrompts and the synthetic streams, 50 for TruthfulQA). Lower is better and the target is $\alpha = 0.1$.
- *Maximum local violation*: the largest mean of err_t over any window of 120 steps (60 for TruthfulQA).
- *Mean threshold*: the average of $\min\{q_t, 1\}$, a measure of sharpness; a lower threshold means fewer unnecessary interventions.
- *Intervention rate*: the fraction of steps with $q_t > \tau$, with $\tau = 0.5$ fixed in advance.
- *Frontier gap*: for each stream, Adaptive CRC is run with five windows. This traces a curve of post-shift violation against mean threshold. A method's frontier gap is its post-shift violation minus the value of that curve, linearly interpolated at the method's own mean threshold. A negative gap means the method does better than Adaptive CRC at the same sharpness. When a method's mean threshold lies outside the range spanned by the five windows, the curve is clipped to its nearest end rather than extrapolated, which can favour or disfavour the method depending on the slope of the curve. For the real streams we therefore also report the gap over the streams whose threshold lies inside the range.

Streams. *Synthetic*: five generators, each with three era boundaries within which the regime is fixed or alternates between two regimes every 50 to 80 steps, run with 30 seeds (0 to 29). These are the generators of the earlier version, with a deterministic per-task seed (the earlier script derived the seed from Python's per-process string hash, so its table could not be reproduced). *Real model outputs*: we run Qwen2.5-1.5B-Instruct on CPU with one fixed prompt template for each task and record the probability the model gives to each answer option. For MMLU (Hendrycks et al., 2021) we use 45 randomly chosen questions per subject (a fixed seed), 2565 in all, and for TruthfulQA (Lin et al., 2021) all 817 questions in the multiple-choice setting, scoring each answer choice by the summed log-probability of its text. The risk is $1 - p(\mathrm{correct}) \in [0, 1]$, and as a second risk we use the 0/1 error. Features are unit-norm all-MiniLM-L6-v2 sentence embeddings

of the question text (Reimers and Gurevych, 2019). For RealToxicityPrompts (Gehman et al., 2020) we draw 4000 of the 99,442 prompts at random (a fixed seed), continue each with GPT-2 (one nucleus sample with $p = 0.9$ and 20 new tokens), and score the continuation alone with the `unitary/toxic-bert` toxicity output; the features are embeddings of the prompt text, and the risk is that toxicity probability. A replication uses Qwen2.5-0.5B continuations scored by `unitary/unbiased-toxic-roberta` on the same prompts. The *orderings* use only the benchmark's own metadata or the question text, never the model's output: random orders of the four broad MMLU categories (24 permutations), the 24 permutations of the four quartile bands of the prompt toxicity annotation (RealToxicityPrompts; the annotation describes the prompt, not the model's output), ascending and descending prompt toxicity and of the 57 subjects (20 orders), random orders of the TruthfulQA categories (20), embedding-direction orders (first principal component, forward and reversed), question-length order, and 20 random permutations as a control with no shift.

Pre-declared settings. All methods use $\alpha = 0.1$, the unprojected update with γ fixed per experiment (0.025 for the synthetic pilot; 0.02 for the other synthetic streams, MMLU and RealToxicityPrompts; 0.03 for TruthfulQA, which has $n = 817$), the correction factor $1+m_t$, and a pool of $3W$ steps for every weighted rule. On the real streams the calibration prefix and window are 256 with a recent window of 128 (MMLU) and 100 with 50 (TruthfulQA). The kernel bandwidth is the median pairwise distance h among the first prefix prompts of that stream, the gate centre is $1.1h$ and the gate slope is $2.5/h$. These rules were written down before any real-stream result was computed. The synthetic-stream constants are those of the earlier scripts and were not changed; the earlier version chose them while developing the generators (Section 8). The analyses added after the first review (MMLU at further levels, the kernel bandwidth sweep, the recurring-regime orderings, the test of the drift and the comparison with L-ARC) were written down with their decision rules before they were run, in a time-stamped file in the repository linked on acceptance. Where the outcome did not fit a sentence we had written in advance, we say so in the text.

Uncertainty. For synthetic streams we report the mean over 30 seeds with the standard deviation and compare methods by paired differences within seed (mean difference, standard error, and the number of seeds in which one method is lower). For the real streams, the unit of replication is the ordering. For a single ordered stream, standard errors of coverage use a moving-block bootstrap of the violation indicator with block length equal to the recent window.

6 Results

6.1 Synthetic streams

Each of the five synthetic generators has three era boundaries at which the prompt features and the risk change; within an era the regime is fixed or alternates between two regimes every 50 to 80 steps. Table 1 reports the mean over 30 seeds, and Figure 1 plots post-shift violation against sharpness.

The update does the long-run work. Static calibration reaches coverage 0.334 and rolling calibration 0.854, below the target of 0.90. Every method with the online update is within 0.02 of the target, consistent with Theorem 2 (the bound is 0.018 to 0.024 for these streams and is listed for every experiment in Table 36). Removing the update from PromptShift-CRC lowers its coverage

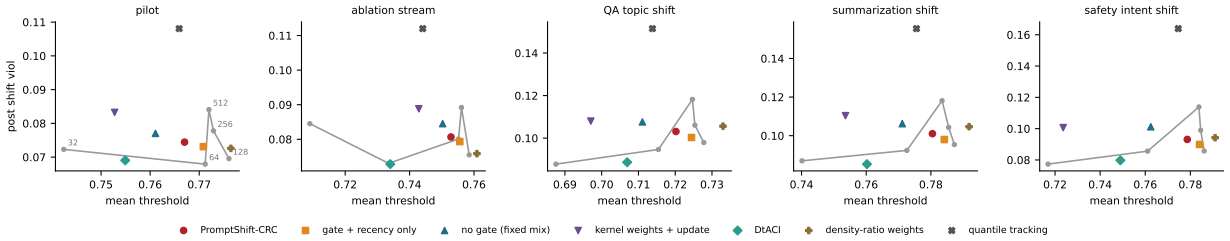


Figure 1: Validity and sharpness on the five synthetic streams (mean over 30 seeds). The grey curve is Adaptive CRC with windows 32, 64, 128, 256, 512 (labelled on the first panel); points are other methods. Lower is better on both axes.

to 0.879 and raises its post-shift violation from 0.090 to 0.214. Marginal coverage therefore does not separate the weighting schemes, and we compare them on post-shift violation, maximum local violation and sharpness.

PromptShift-CRC against the matched baseline. Adaptive CRC with a window equal to PromptShift-CRC's recent window has post-shift violation 0.088, against 0.090 for PromptShift-CRC. Paired by seed (Table 9), PromptShift-CRC minus the matched baseline is between +0.0016 and +0.0032 across the five streams, and PromptShift-CRC is lower in only 0 to 3 of 30 seeds, so it is slightly *worse* on local validity. It is sharper: the difference in mean threshold is between -0.0058 and -0.0031, and PromptShift-CRC is lower in 30 to 30 of 30 seeds. The two differences have opposite signs and are of the same size, about half a percentage point, so the comparison is a trade-off and not a win.

What the components do. The gate buys validity at a cost in sharpness. PromptShift-CRC minus the variant with no gate has post-shift violation between -0.0081 and -0.0026 (lower in 23 to 30 of 30 seeds) and threshold between +0.0026 and +0.0162. The kernel weights do the opposite. The variant that keeps only the gate and the recency window has post-shift violation 0.088 and mean threshold 0.764, against 0.090 and 0.760 for PromptShift-CRC. The kernel weights with the update but without the gate and the recency fallback have the sharpest thresholds among these variants (0.734) and the highest post-shift violation (0.098).

Frontier gap. A fairer comparison asks whether a method does better than Adaptive CRC at the same sharpness. Table 2 gives the frontier gap. For PromptShift-CRC it ranges from -0.0055 to +0.0015 over the five streams (standard error at most 0.0019) and is negative on 3 of the five streams. No stream shows more than about half a percentage point of post-shift violation recovered by the drift-aware components.

Stronger online methods. DtACI (Gibbs and Candès, 2024), which runs several step sizes and weights them online, has post-shift violation 0.079 on average, and PromptShift-CRC minus DtACI is between +0.0054 and +0.0158 on the five streams, with PromptShift-CRC higher in every stream. Its threshold is also lower. Part of this comes from over-coverage: DtACI's coverage is 0.909 on average, above the target, which flatters its local violation, and the frontier gap is the fairer view. Quantile tracking is worse than every weighted rule that uses the update on these streams. The density-ratio rule is close to PromptShift-CRC. Among the online methods we tested, the handling of the step size matters about as much as the choice of weights.

6.2 Real model outputs

We ran three open models on CPU and recorded their outputs (Section 5): Qwen2.5-1.5B-Instruct for MMLU and TruthfulQA, GPT-2 for RealToxicityPrompts, and Qwen2.5-0.5B for the replication. The risk is $1 - p(\text{correct})$ for the two question-answering tasks and the toxicity probability of a continuation for the third. The tasks differ in one way that matters for this paper: how much the *upper tail* of the risk moves between regimes, because the threshold q_t targets the 0.9 quantile.

MMLU and TruthfulQA: the upper tail does not move. Qwen2.5-1.5B-Instruct answers 0.605 of the 2565 MMLU questions correctly. Across the four broad categories the median risk moves a great deal (from 0.097 to 0.551) but the 0.9 quantile does not (0.955 to 0.980). A threshold at the 0.9 quantile is therefore close to 1 in every regime, and the methods tie (Table 10). In the category-block orderings static calibration already has post-shift violation 0.097, against 0.098 for PromptShift-CRC and 0.099 for matched Adaptive CRC; the mean thresholds of the online methods lie within 0.015 of one another; and the intervention rate is 0.999 or higher for every method, so the decision rule almost never lets the model answer. PromptShift-CRC minus matched Adaptive CRC is -0.0010 (SE 0.0007) in post-shift violation.

The frontier gaps are -0.0007 (SE 0.0006) for PromptShift-CRC and -0.0084 (SE 0.0012) for DtACI in the category-block orderings, and -0.0001 and -0.0060 in the subject-block orderings. In 7 of the 24 category-block orderings the mean threshold of PromptShift-CRC lies outside the range of the window sweep, where the frontier is clipped to its nearest end; over the 17 orderings inside the range the PromptShift-CRC gap is +0.0001 (SE 0.0006), against -0.0007 over all 24.

On TruthfulQA the model answers 0.237 of the questions correctly under multiple-choice scoring, the 0.9 quantile of the risk is 1.000, and the same holds. At this level these two tasks test the guarantee (every online method reaches the target) but not the weighting, because no weighting can help when the quantity being estimated barely changes. We report them so that the reader can see the setting in which drift-awareness has nothing to offer.

MMLU at levels where the target quantile moves. The target quantile moves more at higher levels. For the category blocks the range over the four categories of the $(1 - \alpha)$ quantile of the risk is 0.025 at $\alpha = 0.1$, 0.077 at $\alpha = 0.2$ and 0.242 at $\alpha = 0.3$ (the 0.7 quantile runs from 0.54 to 0.78), so MMLU at $\alpha = 0.3$ is a second real task on which the target quantile moves. We ran the category blocks at these two further levels with every constant as declared, and wrote down before the runs that any effect of the drift-aware components should grow with the movement of the target. It does not (Table 32). PromptShift-CRC minus the matched baseline in post-shift violation is -0.0016 (SE 0.0009) at $\alpha = 0.2$ and -0.0004 (SE 0.0013) at $\alpha = 0.3$, and the frontier gap of PromptShift-CRC is +0.0003 (SE 0.0007) and +0.0002 (SE 0.0012), against -0.0007 at $\alpha = 0.1$. At $\alpha = 0.3$ the gap over the 23 orderings inside the range of the window sweep is +0.0004 (SE 0.0013). The criterion we had fixed for claiming that the components help at a moving target (a gap of at most -0.005, and more than two standard errors, at $\alpha = 0.3$) is not met. As on RealToxicityPrompts, the components move post-shift violation by less than half a percentage point at equal sharpness, so a moving target is not sufficient for these weights to help.

RealToxicityPrompts: the upper tail moves. We drew 4000 prompts, continued each with GPT-2 (one nucleus sample of 20 tokens), and scored the continuation with a public toxicity classifier. Only 0.041 of the continuations have risk above 0.5, and they are concentrated in prompts that are themselves toxic. Split by quartiles of the prompt toxicity annotation, the 0.9 quantile of the risk is 0.004 in the lowest band and 0.583 in the highest, so a change of band is a real shift of

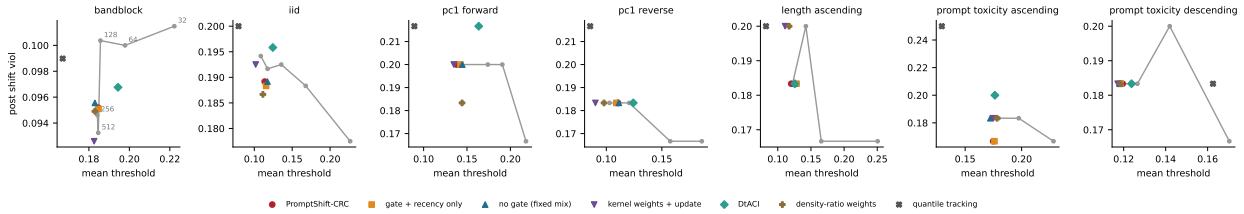


Figure 2: Validity and sharpness on RealToxicityPrompts with GPT-2 continuations. Panels are ordering families (each point is the mean over the orderings in the family). The grey curve is Adaptive CRC over the window sweep; labels give the window.

the target quantile. The ordering uses only the annotation of the prompt, not the model's output. In the three lowest bands the 0.9 quantile lies below 0.06, close to the classifier's noise floor, so the intervention rate at $\tau = 0.5$ carries little information on this task and we do not interpret it.

Static calibration fails here (coverage 0.867, post-shift violation 0.144), and rolling calibration is better but still above target after shifts (0.114). All online methods reach the target coverage (Table 16, Figure 2). Among them the differences are small. Post-shift violation is 0.095 for PromptShift-CRC, 0.100 for the matched baseline, 0.095 for Adaptive CRC with the window W and 0.093 with the longest window of the sweep. Paired by ordering, PromptShift-CRC minus the matched baseline is -0.0052 (SE 0.0011, PromptShift-CRC lower in 19 of 24 orderings). On this stream a longer window works about as well as PromptShift-CRC, which is the opposite of the synthetic streams, where short windows were best. The frontier gap is -0.0010 (SE 0.0010; -0.0012, SE 0.0013, over the 19 of 24 orderings inside the range of the sweep) for PromptShift-CRC, -0.0027 (SE 0.0012) for the kernel weights with the update, and -0.0030 (SE 0.0015) for DtACI. The kernel with the median-distance bandwidth barely localises on sentence embeddings (the effective size of the kernel rule is close to its pool length), so its behaviour is close to that of uniform weights; this is a property of the rule with that bandwidth and we did not tune it.

Replication with a second generator and classifier. We repeated the experiment on the same 4000 prompts and the same bands, with continuations from Qwen2.5-0.5B scored by a different classifier (`unitary/unbiased-toxic-roberta`). The risks of the two runs are only weakly related (correlation 0.20), because the continuations differ. Here 0.037 of the continuations have risk above 0.5, and the 0.9 quantile of the risk is 0.004 in the lowest band and 0.446 in the highest, so the upper tail moves again. The results agree with those above (Table 31). Static calibration fails (coverage 0.863, post-shift violation 0.144); post-shift violation is 0.092 for PromptShift-CRC, 0.097 for the matched baseline, 0.092 for Adaptive CRC with the window W and 0.090 for the kernel weights with the update. PromptShift-CRC minus the matched baseline is -0.0050 (SE 0.0015, lower in 16 of 24 orderings) and its frontier gap is -0.0003 (SE 0.0007); the frontier gap of DtACI is -0.0019 (SE 0.0015). As on the first generator, a longer window does as well as PromptShift-CRC.

Other target levels. Running the GPT-2 streams at $\alpha = 0.05$ and $\alpha = 0.2$ gives the same pattern (Table 31). PromptShift-CRC minus the matched baseline in post-shift violation is -0.0039 (SE 0.0009) at $\alpha = 0.05$ and -0.0031 (SE 0.0019) at $\alpha = 0.2$; at $\alpha = 0.2$ the difference is not distinguishable from zero. The frontier gap of PromptShift-CRC is -0.0010 (SE 0.0009) at $\alpha = 0.05$ and -0.0017 (SE 0.0015) at $\alpha = 0.2$.

6.3 Robustness to the window choice

The results suggest that the value of PromptShift-CRC is that no window has to be chosen. The best window for Adaptive CRC depends on the stream. It is short on the synthetic streams and long on RealToxicityPrompts, and the window that is best in hindsight is known only afterwards. We state the comparison in two ways.

Rank. Among the five Adaptive CRC windows and PromptShift-CRC, PromptShift-CRC has mean rank 3.9 on the synthetic streams, 3.2 and 3.2 on MMLU, 3.0 on RealToxicityPrompts and 3.9 on TruthfulQA (Table 5; 1 is the lowest violation and 3.5 is the middle of six). It is in the better half on MMLU and RealToxicityPrompts, and not on the synthetic streams or on TruthfulQA. In every setting it is better than the worst window, by between two-fifths and two-thirds of the spread of the family.

Choosing the window elsewhere. The best window in hindsight is selected on the data it is scored on, which biases the comparison against PromptShift-CRC. A fairer protocol chooses the window on the other families of streams and applies it to the held-out family (Table 6). PromptShift-CRC minus the chosen window is -0.0030 (synthetic), -0.0019 and -0.0006 (MMLU), -0.0048 (RealToxicityPrompts) and +0.0014 (TruthfulQA). It is lower in four of the five families (Table 6 gives standard errors): clearly so on the synthetic streams and on RealToxicityPrompts, within two standard errors on the two MMLU families, and it loses on TruthfulQA (+0.0014, SE 0.0006). The best window changes from the shortest (synthetic) to the longest (RealToxicityPrompts), and that change is the argument for a rule that does not choose a window. We do not claim that PromptShift-CRC beats a well-chosen window; the evidence is that it avoids a poorly chosen one.

6.4 What each component does

On the synthetic streams and on RealToxicityPrompts the same pattern holds (Tables 1 and 16). The online update is the component that matters most: removing it moves post-shift violation from 0.090 to 0.214 on the synthetic streams and from 0.095 to 0.112 on RealToxicityPrompts. The recency window with the gate is next: the variant that keeps only the gate and the recency window has post-shift violation 0.088 (synthetic) and 0.095 (RealToxicityPrompts), close to PromptShift-CRC's 0.090 and 0.095. The kernel weights contribute least. With them alone and the update, post-shift violation is 0.098 on the synthetic streams, higher than PromptShift-CRC's, with a sharper threshold. These components move post-shift violation by less than one percentage point and the mean threshold by up to 1.6 points, and the sign depends on the stream.

Table 7 puts every real setting with a post-shift analysis on one axis, the range over regimes of the target quantile of the risk. In the 8 settings the range runs from 0.000 to 0.858, and the frontier gap of PromptShift-CRC is at most 0.0020 in absolute value in every one of them, whether it is taken over all orderings or over those inside the range of the window sweep. The largest differences from the matched baseline are on RealToxicityPrompts, where PromptShift-CRC minus the matched baseline in post-shift violation is between -0.0031 and -0.0052 at the three GPT-2 levels, but a longer window does as well, so these are not gains over the frontier. So the data do not show a setting in which the drift-aware components recover more than a fraction of a percentage point of post-shift violation at equal sharpness, whatever the movement of the target quantile.

6.5 Does the computed drift predict violations?

An earlier draft described the post-gate drift D_t as a useful diagnostic. We tested whether the pre-gate drift, which does not depend on the scores, predicts violations, with three statistics fixed before the traces were analysed (Table 35): the violation rate in the steps of a stream whose drift

is in its top decile minus the rate elsewhere, the AUC of the drift for the violation indicator, and, across the shifts of a family, the Spearman correlation between the jump in drift at a boundary and the post-shift violation. We had fixed the criterion for success as a top-decile difference of at least 0.02 (and more than two standard errors) or an AUC of at least 0.55, on at least one real family.

The top-decile criterion is met on one of the three real families, MMLU categories (+0.024, SE 0.0035). The AUC criterion is met on none (0.51 on MMLU, 0.43 and 0.44 on RealToxicityPrompts). By the rule we fixed in advance, which required either criterion on one real family, the drift therefore counts as predictive. We do not describe it as an alarm, because the sign reverses on RealToxicityPrompts, where the AUC is also below 0.5, a case the rule did not anticipate. The result is not a consistent alarm. On RealToxicityPrompts the sign reverses: the steps with the highest drift have fewer violations than the others (-0.044 with GPT-2 and -0.027 with Qwen2.5-0.5B). On all three real families the shift-level correlation is negative (-0.66 to -0.25), so the larger the jump in drift at a boundary, the lower the violation rate after it. On the synthetic streams the step-level association is positive (+0.047 to +0.071), but the shift-level correlation changes sign from stream to stream. The matched baseline gives the same signs. On the real streams the pre-gate drift is almost flat across a boundary (on RealToxicityPrompts 1.354 in the 20 steps before and 1.362 in the 25 steps after), and the violation rate in the first 25 steps after a boundary is 0.121 against 0.090 in the next 125 steps on RealToxicityPrompts, with no excess on MMLU (0.095 and 0.098). We therefore describe the drift as a measure of how unlike its pool a prompt is, and not as an alarm for validity.

6.6 A published localised competitor: L-ARC

Localised adaptive risk control (L-ARC, Zecchin and Simeone, 2024) is the published method closest to PromptShift-CRC: an online threshold that depends on the features through a kernel. We ran it on every stream of the main tables (Table 8). The update and its constants are those of the authors; the features are the same unit-norm embeddings as for every other method, with no rescaling, and the threshold starts from the calibration quantile, as it does for quantile tracking. Before the runs we fixed the authors' defaults (step size $\eta_1 = 1$, kernel length scale 1) as the primary configuration and decided to report three other configurations in full (a step size of 0.1, and a length scale matched to our bandwidth; Table 33). We claim no coverage guarantee for L-ARC in this setting.

With its defaults L-ARC holds the target on the synthetic streams (coverage 0.896 to 0.899) but lies above the Adaptive CRC frontier by +0.018 to +0.036 in post-shift violation on all five streams, where PromptShift-CRC is within 0.0055. With the smaller step it does not reach the target (coverage 0.84 to 0.87) and lies above the frontier by +0.17 to +0.24. On MMLU and TruthfulQA it is on the frontier, as every method is (gaps +0.0001 for category blocks, -0.0002 for subject blocks and -0.0008 for TruthfulQA). On RealToxicityPrompts it is below the frontier with the GPT-2 continuations (-0.0033, SE 0.0033; -0.0109 over the 16 orderings inside the range of the sweep) and above it with the Qwen2.5-0.5B continuations (+0.0064, SE 0.0028), where the gaps of PromptShift-CRC are -0.0010 and -0.0003. L-ARC minus PromptShift-CRC in post-shift violation is +0.0011 (SE 0.0027) and +0.0064 (SE 0.0027) on the two runs, and in mean threshold -0.0016 and -0.0107.

The two sentences we had written in advance for this comparison do not fit the outcome, because on the GPT-2 run both L-ARC and PromptShift-CRC are at or below the frontier. We therefore state it directly. With its authors' defaults L-ARC is not better than PromptShift-CRC against the frontier in a consistent way: it is well above the frontier on the synthetic streams, on it on MMLU and TruthfulQA, below it on one RealToxicityPrompts run and above it on the other.

6.7 Do the weights help when the kernel localises?

The kernel at the median bandwidth is close to uniform (Section 6.2), and every block ordering above visits each regime once, so a past prompt like the current one is rarely in the pool when it would matter. We tested both limits with two designs fixed in advance (Table 34). First, we repeated the band-block and category-block runs with the kernel bandwidth set to $h/4$, $h/2$, h and $2h$, with the gate centre and slope rescaled with it, and report every setting. Second, we built orderings in which blocks of 128 prompts cycle through the regimes (seven visits to each toxicity band, four to each MMLU category), so that the previous visit to a regime is still in the pool of 768 steps, and ran the same four bandwidths on them: on RealToxicityPrompts with both generators and on MMLU at $\alpha = 0.1$, 0.2 and 0.3.

The sharper kernels do localise. On RealToxicityPrompts with GPT-2 the effective size of the kernel-plus-update rule falls from 727 at h to 67 at $h/4$, and on MMLU from 703 to 30; the effective size of PromptShift-CRC falls from 385 to 128, because its gate then falls back on the recent window. In the block orderings the frontier gap of PromptShift-CRC is between -0.0010 and +0.0030 across the four bandwidths and three settings. In the recurring-regime orderings it is within 0.0004 of zero in every setting at every bandwidth, and PromptShift-CRC is within 0.0005 of the matched baseline.

The numerical criterion we had fixed for a positive finding (a frontier gap below -0.005 and more than two standard errors, for a kernel variant at a bandwidth) can be read over all 24 orderings or over those inside the range of the window sweep, and the text we wrote in advance did not say which. Over all orderings it is met in 2 of the 160 cells. Both are the same rule, kernel weights with the update (the variant without the recency fallback coincides with it), at $h/4$ on the GPT-2 block orderings (-0.0074, SE 0.0023). In none of those 24 orderings does the mean threshold lie inside the range of the sweep: it is 0.271, against 0.222 for the shortest window and 0.186 for the matched baseline. The rule is much blunter than any window, so its gap is measured against the end of the curve and is not a comparison at equal sharpness. It does not reappear with the Qwen2.5-0.5B continuations (-0.0031, SE 0.0025), and on MMLU the same variant has the opposite sign (+0.0058, SE 0.0015).

Over the orderings inside the range, in the cells that have at least five of them, the criterion is met in 2 of 146 cells. They are the same variant, the no-gate rule (a fixed half-and-half mix of the kernel weights and the recent window) at $h/4$, on the two RealToxicityPrompts runs: -0.0060 (SE 0.0021) over 11 of 24 orderings with GPT-2, and -0.0051 (SE 0.0021) over 17 of 24 with Qwen2.5-0.5B. Over all 24 orderings the gaps are -0.0050 (SE 0.0015) and -0.0023 (SE 0.0018). This is the only configuration in the study that lies about half a percentage point below the frontier, and we regard it as a lead to test, not as a finding; this is a judgment made after the result, because the rule we fixed in advance had no correction for the 146 comparisons. It is two of 146 comparisons, and at a threshold of two standard errors about three cells would be expected by chance if no variant helped, although the cells are correlated. The two runs use the same prompts and orderings. It does not appear on MMLU or in the recurring-regime orderings. It was not the rule we proposed. In the other cells with at least five orderings inside the range, the gap is smaller than half a percentage point in absolute value.

DtACI, which has no kernel, lies clearly below the frontier on the recurring-regime orderings of RealToxicityPrompts, where its gap is -0.0056 (SE 0.0003) with GPT-2 and -0.0061 (SE 0.0003) with Qwen2.5-0.5B, with all 24 orderings inside the range of the sweep.

Table 1: Synthetic streams, mean over five generators of the 30-seed means (and mean seed-to-seed sd). Every online method has coverage near 0.90 because of the update; local violation and sharpness separate the methods.

Method	Coverage	Post-shift violation	Max local violation	Mean threshold
Static	0.334 ± 0.012	0.802 ± 0.016	0.986	0.450
Rolling	0.854 ± 0.005	0.311 ± 0.015	0.524	0.758
Adaptive CRC, window W	0.901 ± 0.001	0.095 ± 0.005	0.201	0.764
Adaptive CRC, matched window	0.901 ± 0.001	0.088 ± 0.006	0.188	0.764
Kernel weights + update	0.903 ± 0.001	0.098 ± 0.005	0.189	0.734
Kernel weights, no update	0.860 ± 0.007	0.295 ± 0.019	0.527	0.737
PromptShift-CRC	0.901 ± 0.001	0.090 ± 0.005	0.186	0.760
Gate + recency only	0.901 ± 0.001	0.088 ± 0.006	0.187	0.764
No gate (fixed mix)	0.902 ± 0.001	0.095 ± 0.005	0.186	0.751
No recency fallback	0.903 ± 0.001	0.098 ± 0.005	0.189	0.734
No online update	0.879 ± 0.004	0.214 ± 0.015	0.387	0.762
DtACI	0.909 ± 0.003	0.079 ± 0.007	0.158	0.741
Density-ratio weights	0.902 ± 0.001	0.091 ± 0.006	0.198	0.771
Quantile tracking	0.891 ± 0.001	0.138 ± 0.005	0.231	0.755

Post-shift violation is the miss rate over the 150 steps after each regime boundary (target 0.10).

Table 2: Frontier gap on the synthetic streams: post-shift violation minus the Adaptive-CRC window-sweep curve interpolated at the method's own mean threshold, mean over seeds with standard error. Negative means better than Adaptive CRC at the same sharpness.

Stream	PromptShift-CRC	Gate + recency only	Kernel + update	DtACI	Density ratio
pilot	+0.0015 (0.0011)	-0.0023 (0.0009)	+0.0111 (0.0010)	-0.0032 (0.0013)	-0.0002 (0.0012)
ablation stream	-0.0013 (0.0008)	-0.0033 (0.0008)	+0.0113 (0.0010)	-0.0015 (0.0014)	-0.0030 (0.0013)
QA topic shift	+0.0015 (0.0011)	-0.0032 (0.0010)	+0.0179 (0.0013)	-0.0043 (0.0016)	+0.0028 (0.0016)
summarization shift	-0.0049 (0.0019)	-0.0054 (0.0011)	+0.0212 (0.0011)	-0.0050 (0.0013)	+0.0045 (0.0017)
safety-intent shift	-0.0055 (0.0014)	-0.0049 (0.0014)	+0.0218 (0.0009)	-0.0037 (0.0014)	-0.0010 (0.0019)

7 Discussion

What a practitioner should take from this. If the aim is long-run control of a risk threshold under prompt shift, use an online update with a calibration window of the right length. The update gives the guarantee (Theorem 2), and the best window differed between our streams. Among the online rules we tried, DtACI was the strongest on the synthetic streams, on MMLU and on the recurring-regime orderings of RealToxicityPrompts. PromptShift-CRC is a reasonable choice when the right window is unknown: in every setting it did better than the worst window of the Adaptive CRC family, and it was in the better half of the family on MMLU and RealToxicityPrompts. Its computed prompt drift D_t and effective size $n_{\text{eff},t}$ say how far the current prompt is from the evidence behind the threshold. We tested whether the drift predicts violations (Section 6.5) and found that it does not do so reliably, so we describe it as a measure of novelty and not as an alarm.

Why the weights add little. Theorem 2 gives the update full responsibility for the long-run rate. The weights can then only improve the *local* rate and the sharpness. Two things limit what they can do. First, a kernel on sentence embeddings groups prompts by topic, and the risk of a prompt depends only weakly on its topic, so a topic-similar past prompt is a weak guide to the current risk; MMLU is the clearest example, since the median risk differs across categories while the

Table 3: Frontier gap on MMLU streams (post-shift violation minus the Adaptive-CRC window-sweep curve at the same mean threshold; mean over orderings with standard error). Negative is better than Adaptive CRC. Where a method's mean threshold lies outside the range of the window sweep the curve is clipped to its nearest end; the last column repeats the PromptShift-CRC gap over the orderings whose threshold lies inside the range.

Ordering	PromptShift-CRC	Gate + recency only	Kernel + update	DtACI	Density ratio	PromptShift-CRC, inside range
cat blocks	-0.0007 (0.0006)	-0.0007 (0.0006)	-0.0007 (0.0006)	-0.0084 (0.0012)	-0.0006 (0.0006)	+0.0001 (0.0006), 17 of 24
subject blocks	-0.0001 (0.0001)	-0.0001 (0.0001)	+0.0002 (0.0001)	-0.0060 (0.0006)	+0.0001 (0.0002)	-0.0002 (0.0001), 18 of 20

Table 4: Frontier gap on RTP streams (post-shift violation minus the Adaptive-CRC window-sweep curve at the same mean threshold; mean over orderings with standard error). Negative is better than Adaptive CRC. Where a method's mean threshold lies outside the range of the window sweep the curve is clipped to its nearest end; the last column repeats the PromptShift-CRC gap over the orderings whose threshold lies inside the range.

Ordering	PromptShift-CRC	Gate + recency only	Kernel + update	DtACI	Density ratio	PromptShift-CRC, inside range
band blocks	-0.0010 (0.0010)	-0.0007 (0.0011)	-0.0027 (0.0012)	-0.0030 (0.0015)	+0.0006 (0.0008)	-0.0012 (0.0013), 19 of 24

upper tail does not. Second, down-weighting costs effective size (Lemma 1), and with a threshold at the 0.9 quantile that cost is paid in the level of the quantile. On RealToxicityPrompts, where the upper tail does move between bands, a longer effective window helps a little. The kernel at the median bandwidth is close to uniform there: a prompt from the highest toxicity band receives 1.03 times the kernel weight of a prompt from another band, and the effective size of the kernel rule is 727 of a pool of 768. The gain is therefore a window effect and not a similarity effect. A kernel four times sharper does localise, but it does not move post-shift violation by more than about half a percentage point at equal sharpness, with one lead that we do not claim (Section 6.7).

8 Limitations

- **Scale of the real-output experiments.** We used three small open models (Qwen2.5-1.5B-Instruct, GPT-2 and Qwen2.5-0.5B) on CPU, two toxicity classifiers, one prompt template, one sampled continuation per prompt, and one embedding model. Larger models, other templates and other embeddings may behave differently. We generated toxic text only to measure its toxicity, and we release risk scores and not the text. The target quantile moves materially only on RealToxicityPrompts and on MMLU at $\alpha = 0.3$, and neither shows a gain over the frontier; a task with a different kind of risk, such as summarisation faithfulness, would strengthen the conclusions.
- **The synthetic generators and their constants.** The generators and the per-experiment constants (bandwidth, gate centre and slope) are those of the earlier version of this work and were chosen while the generators were being developed, so they are not independent of the generators. We did not tune any constant after seeing the results reported here, and the constants on the real streams follow rules fixed in advance (Section 5).
- **Theory.** Theorem 1 needs conditional independence of the risks given the features and applies at a fixed level; Corollary 1 needs a Lipschitz condition that we cannot check. The long-run result (Theorem 2) holds for any weights but says nothing about local behaviour. Proposition 1 bounds the conditional violation probability at the running level, but we have no guarantee that the running level stays near α over a short window, and the window identity

Table 5: Robustness to the window choice. For each stream, Adaptive CRC is run with a sweep of windows. PromptShift-CRC (PS) has no window to choose. Entries are means over streams of post-shift violation: PS minus the best window in hindsight, the spread of the window family (worst minus best), PS minus the worst window, and the rank of PS among the window family and PS (1 = lowest violation). The best window in hindsight is selected on the same data, so the first column is biased in PS's disfavour.

Setting	Streams	PS − best	Spread	PS − worst	Rank of PS
synthetic (5 streams x 30 seeds)	150	+0.0124	0.0267	-0.0143	3.9 of 6
MMLU, category blocks	24	+0.0039	0.0100	-0.0061	3.2 of 6
MMLU, subject blocks	20	+0.0007	0.0020	-0.0013	3.2 of 6
RealToxicityPrompts, toxicity bands	24	+0.0046	0.0144	-0.0097	3.0 of 6
TruthfulQA, category blocks	20	+0.0051	0.0090	-0.0038	3.9 of 6
RealToxicityPrompts replication (Qwen2.5-0.5B)	24	+0.0043	0.0149	-0.0106	3.0 of 6

Table 6: Choosing the Adaptive CRC window without looking at the held-out family. For each family of streams, the window (position in the five-window sweep, short to long) with the best mean rank over the other four families is chosen; the entry is PromptShift-CRC's post-shift violation minus that window's on the held-out family, with the paired standard error over streams or seeds (negative: PromptShift-CRC is lower). The last two columns give the window that is best on the held-out family itself and PromptShift-CRC's gap to that window in hindsight.

Held-out family	Window chosen elsewhere	PS − chosen	Best here	PS − best here
synthetic	4 of 5	-0.0030 (0.0003)	1 of 5	+0.0124
MMLU categories	2 of 5	-0.0019 (0.0011)	4 of 5	+0.0039
MMLU subjects	2 of 5	-0.0006 (0.0004)	4 of 5	+0.0007
RTP bands	2 of 5	-0.0048 (0.0015)	5 of 5	+0.0046
TruthfulQA categories	4 of 5	+0.0014 (0.0006)	2 of 5	+0.0051

in Remark 1 shows that this is what the local violation rate measures.

- **Proxy streams.** The streams built from TruthfulQA, RealToxicityPrompts, SummEval and LongHalQA fields in the earlier version contain no model outputs and some are ordered by a field linked to the outcome. We report them in Appendix C and draw no conclusion from them.
- **Uncertainty on the real streams.** Standard errors on the real streams are over orderings of one fixed set of items, so they measure ordering variability only, not variability across item sets or generations. The released risk scores are canonical: sampled continuations are not bit-reproducible across hardware.
- **The toxicity task.** The toxicity classifier is the ground truth, and 81% of the risks lie below 0.01, close to its floor; the 0.9 quantile in the three lowest bands is in that range. Only abrupt block shifts have known boundaries, so post-shift violation is reported for them; smooth orderings (embedding direction, length) are reported with local violation only.
- **Weak localisation.** With the bandwidth set to the median pairwise distance, the kernel weights are close to uniform on all of the real streams (effective size 727 of 768 on RealToxicityPrompts and 703 on MMLU), not only on RealToxicityPrompts, which limits what "the kernel weights contribute least" can mean. Sharper kernels do localise (Section 6.7), and we report them in full. TruthfulQA multiple-choice accuracy of the small model is near chance, so that task is uninformative about weighting.

Table 7: How far the target quantile moves between regimes, and what the drift-aware components recover. The spread is the range over regimes with at least 30 prompts of the $(1-\alpha)$ quantile of the risk (a property of the data and the regime labels); the last three columns are PromptShift-CRC minus the matched baseline in post-shift violation, the frontier gap, and the frontier gap over the orderings inside the range of the window sweep.

Setting	Level α	Spread of target quantile	PS − matched post-shift	Frontier gap PS	PS, inside range
RealToxicityPrompts, GPT-2	0.05	0.858	-0.0039 (0.0009)	-0.0010 (0.0009)	-0.0017 (0.0009)
RealToxicityPrompts, GPT-2	0.1	0.579	-0.0052 (0.0011)	-0.0010 (0.0010)	-0.0012 (0.0013)
RealToxicityPrompts, GPT-2	0.2	0.094	-0.0031 (0.0019)	-0.0017 (0.0015)	-0.0020 (0.0015)
RealToxicityPrompts, Qwen2.5-0.5B	0.1	0.442	-0.0050 (0.0015)	-0.0003 (0.0007)	-0.0007 (0.0008)
MMLU	0.1	0.025	-0.0010 (0.0007)	-0.0007 (0.0006)	+0.0001 (0.0006)
MMLU	0.2	0.077	-0.0016 (0.0009)	+0.0003 (0.0007)	+0.0007 (0.0007)
MMLU	0.3	0.242	-0.0004 (0.0013)	+0.0002 (0.0012)	+0.0004 (0.0013)
TruthfulQA	0.1	0.000	+0.0013 (0.0009)	-0.0004 (0.0007)	-0.0004 (0.0007)

Table 8: L-ARC with the authors' defaults against the Adaptive CRC frontier and against PromptShift-CRC (PS). Frontier gap: post-shift violation minus the window-sweep curve at the method's own mean threshold (negative is better), with the gap over the orderings inside the range of the sweep where there are enough of them. Last two columns: L-ARC minus PromptShift-CRC in post-shift violation (standard error) and in mean threshold. Synthetic: mean over the five streams of the 30-seed means. No coverage guarantee is claimed for L-ARC here.

Setting	Gap of PS	Gap of L-ARC	L-ARC, inside range	L-ARC − PS post-shift	L-ARC − PS threshold
synthetic (mean of the five streams)	-0.0017	+0.0277 (0.0005)	–	+0.0237 (0.0004)	-0.0069
MMLU, category blocks	-0.0007	+0.0001 (0.0010)	+0.0013 (0.0022), 8 of 24	+0.0019 (0.0011)	-0.0091
MMLU, subject blocks	-0.0001	-0.0002 (0.0004)	-0.0002 (0.0004), 20 of 20	+0.0002 (0.0002)	-0.0087
RealToxicityPrompts, GPT-2	-0.0010	-0.0033 (0.0033)	-0.0109 (0.0028), 16 of 24	+0.0011 (0.0027)	-0.0016
RealToxicityPrompts, Qwen2.5-0.5B	-0.0003	+0.0064 (0.0028)	-0.0079 (0.0050), 6 of 24	+0.0064 (0.0027)	-0.0107
TruthfulQA, category blocks	-0.0004	-0.0008 (0.0011)	–	-0.0017 (0.0013)	-0.0044

- **Baselines.** The density-ratio rule is a simplified stand-in for published methods, and L-ARC (Zecchin and Simeone, 2024) was run with the authors' update and defaults but without their feature rescaling and with a warm start (Section 6.6).

9 Conclusion

We set out to show that drift-aware weighting improves conformal risk control for foundation models under prompt shift. What we found is narrower. The online update provides the long-run guarantee for any weights; the weights and the gate change local validity by less than one percentage point and sharpness by up to 1.6 points; a matched short window and DtACI are as good or better in most of our settings; and PromptShift-CRC performs like a mid-ranked window without having to choose one. Tests fixed in advance to give the weights a better chance (a sharper kernel, regimes that recur, a second moving target, a published localised competitor) show no consistent gain over the frontier. We state a conditional bound with a computable prompt-drift term, give an evaluation protocol for comparing such rules fairly, and release the code, the model outputs and the streams so that others can test the drift-aware idea on tasks where the upper tail of the risk moves with the prompt.

Declaration of AI use

The authors used large language models (Anthropic's Claude models) to write and debug code, to run and check the experiments and proofs, and to draft and edit text. The study design, including the evaluation protocol, was developed with the model's assistance; the authors approved each design choice, reviewed every result and every proof, and take full responsibility for the content. No language model is an author. All numbers in the results sections are generated from the released result files by the released scripts.

Data and code availability

The code, the recorded model outputs (risk scores, features and the identifiers needed to rebuild the streams), the result tables and the paper source are in the public repository `https://github.com/jopoku16/promptshift-crc`. Question and prompt text from MMLU and TruthfulQA can be re-downloaded from the public datasets; the RealToxicityPrompts prompt and continuation text is withheld from the release because it contains offensive language, and the generation script regenerates it. Sampled continuations are not bit-reproducible across hardware, so the released risk scores are canonical. Appendix D gives the commands.

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A Weighted quantile uncertainty

Proposition 2 (Weighted quantile uncertainty)**.** *Conditional on $E_{1:t}$, suppose $R_1, \dots, R_{t-1}$ are independent with a common continuous distribution function F (the no-drift case). For any $\eta \in (0,1)$ and integer $M \geq 1$, with probability at least $1-\eta$,*

$$\sup_x \Big| \sum_{s<t} p_{t,s} \mathbf{1}\{R_s \leq x\} - F(x) \Big| \;\leq\; \sqrt{\frac{\log(2M/\eta)}{2\, n_{\mathrm{eff},t}}} + \frac{1}{M}.$$

Table 9: PromptShift-CRC minus a comparator, paired by seed. Negative means PromptShift-CRC is lower (better for violation; sharper for threshold).

Stream	Comparator	Δ post-shift	SE	PS lower in	Δ threshold	PS lower in
pilot	matched Adaptive CRC	+0.0016	0.0004	2/30	-0.0043	30/30
	no gate	-0.0026	0.0004	23/30	+0.0059	1/30
	gate + recency only	+0.0013	0.0004	4/30	-0.0038	30/30
	DtACI	+0.0054	0.0010	4/30	+0.0121	0/30
ablation stream	matched Adaptive CRC	+0.0018	0.0005	3/30	-0.0031	30/30
	no gate	-0.0039	0.0005	26/30	+0.0026	3/30
	gate + recency only	+0.0013	0.0004	3/30	-0.0027	30/30
	DtACI	+0.0079	0.0011	2/30	+0.0189	0/30
QA topic shift	matched Adaptive CRC	+0.0027	0.0004	0/30	-0.0042	30/30
	no gate	-0.0045	0.0007	24/30	+0.0091	0/30
	gate + recency only	+0.0028	0.0004	1/30	-0.0042	30/30
	DtACI	+0.0144	0.0014	0/30	+0.0133	0/30
summarization shift	matched Adaptive CRC	+0.0030	0.0005	2/30	-0.0042	30/30
	no gate	-0.0052	0.0004	29/30	+0.0093	0/30
	gate + recency only	+0.0030	0.0004	1/30	-0.0037	29/30
	DtACI	+0.0158	0.0011	0/30	+0.0203	0/30
safety-intent shift	matched Adaptive CRC	+0.0032	0.0004	0/30	-0.0058	30/30
	no gate	-0.0081	0.0005	30/30	+0.0162	0/30
	gate + recency only	+0.0032	0.0004	0/30	-0.0054	30/30
	DtACI	+0.0133	0.0014	2/30	+0.0297	0/30

Proof. Let $u_j = j/M$ and $x_j = F^{-1}(u_j)$ for $j = 1, \dots, M-1$, and $x_0 = -\infty$, $x_M = +\infty$; by continuity $F(x_j) = u_j$. Let $\hat{F}(x) = \sum_s p_{t,s} \mathbf{1}\{R_s \le x\}$. For fixed x_j, $\hat{F}(x_j) - F(x_j)$ is a sum of independent terms $p_{t,s}(\mathbf{1}\{R_s \le x_j\} - F(x_j))$, each in an interval of length $p_{t,s}$, so Hoeffding's inequality gives $\mathbb{P}(|\hat{F}(x_j) - F(x_j)| > \varepsilon) \le 2\exp(-2\varepsilon^2 / \sum_s p_{t,s}^2) = 2\exp(-2\varepsilon^2 n_{\mathrm{eff},t})$. A union bound over $j = 1, \dots, M-1$ with $\varepsilon = \sqrt{\log(2M/\eta)/(2n_{\mathrm{eff},t})}$ makes all of them hold with probability at least $1 - \eta$. Because F is continuous, almost surely no R_s equals any x_j, so $\hat{F}(x_j^-) = \hat{F}(x_j)$. For $x \in [x_j, x_{j+1})$, monotonicity gives $\hat{F}(x) - F(x) \le \hat{F}(x_{j+1}^-) - F(x_j) \le \varepsilon + F(x_{j+1}) - F(x_j) = \varepsilon + 1/M$ and $\hat{F}(x) - F(x) \ge \hat{F}(x_j) - F(x_{j+1}) \ge -\varepsilon - 1/M$. □

B Settings and additional tables

C Proxy streams without model outputs

Table 30 reports streams built from fields of TruthfulQA, RealToxicityPrompts, SummEval and LongHalQA, as in the earlier version of this work. They contain no model outputs. The features are hand-built summaries of text, the risk is a score derived from annotations in the benchmark, and some orderings use a field linked to the outcome (for example the human consistency rating in SummEval). The two repeated tail segments of the earlier SummEval and LongHalQA streams have been removed. All online methods reach the target coverage to within the standard error. We draw no conclusion from these streams about the value of drift-aware weighting.

Table 10: MMLU with Qwen2.5-1.5B-Instruct, category blocks (24 streams; mean and sd over streams). Risk: one minus the probability of the correct answer.

Method	Coverage	Post-shift violation	Max local violation	Mean threshold	Intervention rate
Static	0.900 ± 0.022	0.097 ± 0.026	0.183	0.964	1.000
Rolling	0.903 ± 0.002	0.091 ± 0.011	0.167	0.966	1.000
Adaptive CRC, window W	0.900 ± 0.001	0.097 ± 0.009	0.146	0.961	1.000
Adaptive CRC, matched window	0.900 ± 0.001	0.099 ± 0.009	0.148	0.960	1.000
Kernel weights + update	0.900 ± 0.001	0.097 ± 0.009	0.146	0.961	1.000
Kernel weights, no update	0.902 ± 0.005	0.094 ± 0.011	0.177	0.965	1.000
PromptShift-CRC	0.900 ± 0.001	0.098 ± 0.009	0.146	0.961	1.000
Gate + recency only	0.900 ± 0.001	0.098 ± 0.009	0.146	0.961	1.000
No gate (fixed mix)	0.900 ± 0.001	0.098 ± 0.009	0.146	0.961	1.000
No recency fallback	0.900 ± 0.001	0.097 ± 0.009	0.146	0.961	1.000
No online update	0.903 ± 0.003	0.093 ± 0.012	0.165	0.966	1.000
DtACI	0.907 ± 0.002	0.090 ± 0.010	0.150	0.966	1.000
Density-ratio weights	0.900 ± 0.001	0.097 ± 0.008	0.144	0.961	1.000
Quantile tracking	0.900 ± 0.001	0.098 ± 0.006	0.137	0.960	1.000

Table 11: MMLU with Qwen2.5-1.5B-Instruct, subject blocks (20 streams; mean and sd over streams). Risk: one minus the probability of the correct answer.

Method	Coverage	Post-shift violation	Max local violation	Mean threshold	Intervention rate
Static	0.905 ± 0.029	0.094 ± 0.030	0.246	0.966	1.000
Rolling	0.901 ± 0.004	0.099 ± 0.005	0.227	0.961	1.000
Adaptive CRC, window W	0.900 ± 0.001	0.100 ± 0.002	0.160	0.951	1.000
Adaptive CRC, matched window	0.900 ± 0.001	0.100 ± 0.002	0.161	0.950	1.000
Kernel weights + update	0.900 ± 0.001	0.100 ± 0.002	0.157	0.953	1.000
Kernel weights, no update	0.903 ± 0.007	0.098 ± 0.008	0.235	0.964	1.000
PromptShift-CRC	0.900 ± 0.001	0.100 ± 0.002	0.156	0.951	1.000
Gate + recency only	0.900 ± 0.001	0.100 ± 0.002	0.156	0.952	1.000
No gate (fixed mix)	0.900 ± 0.001	0.100 ± 0.002	0.156	0.951	1.000
No recency fallback	0.900 ± 0.001	0.100 ± 0.002	0.157	0.953	1.000
No online update	0.903 ± 0.006	0.096 ± 0.007	0.209	0.963	1.000
DtACI	0.906 ± 0.003	0.094 ± 0.003	0.157	0.956	1.000
Density-ratio weights	0.900 ± 0.001	0.100 ± 0.002	0.157	0.952	1.000
Quantile tracking	0.900 ± 0.001	0.100 ± 0.002	0.147	0.951	1.000

D Reproduction

All experiments run on a CPU. The model outputs take about three hours (MMLU with a 1.5B-parameter model), forty minutes (TruthfulQA), an hour (RealToxicityPrompts with GPT-2) and about two hours (the Qwen2.5-0.5B replication); the stream analyses take one to two hours each on an eight-core machine. The commands are in the repository `README`: `gen_mmlu_outputs.py`, `gen_truthfulqa_outputs.py`, `gen_rtp_outputs.py`, then `run_phase_a_synthetic.py`, `run_phase_a_baselines.p` `run_real_streams.py`, `analyze_frontier.py`, `analyze_real.py`, `make_tables.py` and `make_figures.py`. The tables and numbers in this paper are produced by `make_tables.py` from the result files.

Table 12: MMLU with Qwen2.5-1.5B-Instruct, random order, no shift (20 streams; mean and sd over streams). Risk: one minus the probability of the correct answer.

Method	Coverage	Max local violation	Mean threshold	Intervention rate
Static	0.901 ± 0.017	0.180	0.964	1.000
Rolling	0.903 ± 0.002	0.172	0.966	1.000
Adaptive CRC, window W	0.900 ± 0.001	0.147	0.961	1.000
Adaptive CRC, matched window	0.900 ± 0.000	0.146	0.960	1.000
Kernel weights + update	0.900 ± 0.001	0.145	0.962	1.000
Kernel weights, no update	0.901 ± 0.004	0.181	0.965	1.000
PromptShift-CRC	0.900 ± 0.000	0.145	0.961	1.000
Gate + recency only	0.900 ± 0.000	0.145	0.961	1.000
No gate (fixed mix)	0.900 ± 0.000	0.144	0.961	1.000
No recency fallback	0.900 ± 0.001	0.145	0.962	1.000
No online update	0.903 ± 0.002	0.167	0.966	1.000
DtACI	0.906 ± 0.001	0.146	0.965	1.000
Density-ratio weights	0.900 ± 0.001	0.145	0.961	1.000
Quantile tracking	0.900 ± 0.001	0.138	0.960	1.000

Table 13: MMLU with Qwen2.5-1.5B-Instruct, embedding-direction order (forward) (1 stream; mean and sd over streams). Risk: one minus the probability of the correct answer.

Method	Coverage	Max local violation	Mean threshold	Intervention rate
Static	0.902	0.200	0.965	1.000
Rolling	0.900	0.183	0.963	1.000
Adaptive CRC, window W	0.900	0.158	0.960	1.000
Adaptive CRC, matched window	0.900	0.167	0.959	1.000
Kernel weights + update	0.900	0.150	0.962	1.000
Kernel weights, no update	0.896	0.192	0.962	1.000
PromptShift-CRC	0.900	0.150	0.962	1.000
Gate + recency only	0.900	0.150	0.961	1.000
No gate (fixed mix)	0.900	0.150	0.961	1.000
No recency fallback	0.900	0.150	0.962	1.000
No online update	0.897	0.183	0.962	1.000
DtACI	0.903	0.150	0.962	1.000
Density-ratio weights	0.900	0.150	0.960	1.000
Quantile tracking	0.900	0.150	0.959	1.000

Table 14: MMLU with Qwen2.5-1.5B-Instruct, embedding-direction order (reversed) (1 stream; mean and sd over streams). Risk: one minus the probability of the correct answer.

Method	Coverage	Max local violation	Mean threshold	Intervention rate
Static	0.952	0.108	0.987	1.000
Rolling	0.902	0.175	0.963	1.000
Adaptive CRC, window W	0.899	0.167	0.956	1.000
Adaptive CRC, matched window	0.900	0.150	0.954	1.000
Kernel weights + update	0.899	0.150	0.956	1.000
Kernel weights, no update	0.913	0.167	0.967	1.000
PromptShift-CRC	0.899	0.150	0.957	1.000
Gate + recency only	0.899	0.150	0.958	1.000
No gate (fixed mix)	0.899	0.150	0.957	1.000
No recency fallback	0.899	0.150	0.956	1.000
No online update	0.906	0.158	0.965	1.000
DtACI	0.904	0.175	0.961	1.000
Density-ratio weights	0.899	0.150	0.957	1.000
Quantile tracking	0.900	0.142	0.956	1.000

Table 15: MMLU with Qwen2.5-1.5B-Instruct, question-length order (1 stream; mean and sd over streams). Risk: one minus the probability of the correct answer.

Method	Coverage	Max local violation	Mean threshold	Intervention rate
Static	0.921	0.158	0.976	1.000
Rolling	0.905	0.183	0.965	1.000
Adaptive CRC, window W	0.900	0.158	0.958	1.000
Adaptive CRC, matched window	0.899	0.150	0.955	1.000
Kernel weights + update	0.899	0.158	0.958	1.000
Kernel weights, no update	0.900	0.183	0.964	1.000
PromptShift-CRC	0.899	0.150	0.956	1.000
Gate + recency only	0.899	0.150	0.957	1.000
No gate (fixed mix)	0.899	0.150	0.957	1.000
No recency fallback	0.899	0.158	0.958	1.000
No online update	0.902	0.183	0.965	1.000
DtACI	0.906	0.167	0.962	1.000
Density-ratio weights	0.900	0.158	0.957	1.000
Quantile tracking	0.900	0.150	0.958	1.000

Table 16: RealToxicityPrompts with GPT-2 continuations, prompt-toxicity bands in blocks (24 streams; mean and sd over streams). Risk: toxicity probability of the continuation.

Method	Coverage	Post-shift violation	Max local violation	Mean threshold	Intervention rate
Static	0.867 ± 0.081	0.144 ± 0.102	0.390	0.184	0.250
Rolling	0.901 ± 0.005	0.114 ± 0.028	0.297	0.172	0.204
Adaptive CRC, window W	0.900 ± 0.000	0.095 ± 0.011	0.208	0.184	0.180
Adaptive CRC, matched window	0.900 ± 0.000	0.100 ± 0.008	0.207	0.186	0.174
Kernel weights + update	0.900 ± 0.000	0.093 ± 0.012	0.206	0.183	0.179
Kernel weights, no update	0.898 ± 0.012	0.131 ± 0.037	0.348	0.155	0.155
PromptShift-CRC	0.900 ± 0.000	0.095 ± 0.010	0.204	0.185	0.179
Gate + recency only	0.900 ± 0.000	0.095 ± 0.010	0.204	0.185	0.178
No gate (fixed mix)	0.900 ± 0.000	0.096 ± 0.009	0.206	0.183	0.175
No recency fallback	0.900 ± 0.000	0.093 ± 0.012	0.206	0.183	0.179
No online update	0.904 ± 0.008	0.112 ± 0.027	0.290	0.161	0.164
DtACI	0.906 ± 0.001	0.097 ± 0.011	0.217	0.194	0.199
Density-ratio weights	0.900 ± 0.000	0.095 ± 0.010	0.204	0.183	0.177
Quantile tracking	0.900 ± 0.005	0.099 ± 0.017	0.244	0.167	0.151

Table 17: RealToxicityPrompts with GPT-2 continuations, random order, no shift (20 streams; mean and sd over streams). Risk: toxicity probability of the continuation.

Method	Coverage	Max local violation	Mean threshold	Intervention rate
Static	0.906 ± 0.023	0.211	0.096	0.000
Rolling	0.902 ± 0.002	0.224	0.080	0.000
Adaptive CRC, window W	0.900 ± 0.000	0.192	0.118	0.032
Adaptive CRC, matched window	0.900 ± 0.000	0.193	0.136	0.052
Kernel weights + update	0.900 ± 0.000	0.192	0.102	0.016
Kernel weights, no update	0.903 ± 0.003	0.218	0.072	0.000
PromptShift-CRC	0.900 ± 0.000	0.189	0.114	0.028
Gate + recency only	0.900 ± 0.000	0.188	0.116	0.029
No gate (fixed mix)	0.900 ± 0.000	0.189	0.117	0.031
No recency fallback	0.900 ± 0.000	0.192	0.102	0.016
No online update	0.904 ± 0.002	0.210	0.079	0.000
DtACI	0.907 ± 0.001	0.196	0.124	0.019
Density-ratio weights	0.900 ± 0.000	0.187	0.111	0.024
Quantile tracking	0.900 ± 0.001	0.200	0.079	0.000

Table 18: RealToxicityPrompts with GPT-2 continuations, embedding-direction order (forward) (1 stream; mean and sd over streams). Risk: toxicity probability of the continuation.

Method	Coverage	Max local violation	Mean threshold	Intervention rate
Static	0.684	0.717	0.002	0.000
Rolling	0.894	0.250	0.091	0.000
Adaptive CRC, window W	0.899	0.200	0.140	0.052
Adaptive CRC, matched window	0.899	0.200	0.174	0.092
Kernel weights + update	0.899	0.200	0.135	0.042
Kernel weights, no update	0.885	0.267	0.067	0.000
PromptShift-CRC	0.899	0.200	0.138	0.049
Gate + recency only	0.899	0.200	0.142	0.055
No gate (fixed mix)	0.899	0.200	0.145	0.053
No recency fallback	0.899	0.200	0.135	0.042
No online update	0.891	0.250	0.080	0.000
DtACI	0.905	0.217	0.164	0.048
Density-ratio weights	0.900	0.183	0.144	0.057
Quantile tracking	0.896	0.217	0.089	0.000

Table 19: RealToxicityPrompts with GPT-2 continuations, embedding-direction order (reversed) (1 stream; mean and sd over streams). Risk: toxicity probability of the continuation.

Method	Coverage	Max local violation	Mean threshold	Intervention rate
Static	0.947	0.167	0.283	0.000
Rolling	0.909	0.250	0.091	0.000
Adaptive CRC, window W	0.901	0.183	0.109	0.035
Adaptive CRC, matched window	0.901	0.183	0.121	0.049
Kernel weights + update	0.901	0.183	0.090	0.015
Kernel weights, no update	0.917	0.217	0.099	0.000
PromptShift-CRC	0.901	0.183	0.110	0.034
Gate + recency only	0.901	0.183	0.109	0.033
No gate (fixed mix)	0.901	0.183	0.111	0.037
No recency fallback	0.901	0.183	0.090	0.015
No online update	0.914	0.217	0.097	0.000
DtACI	0.908	0.183	0.124	0.023
Density-ratio weights	0.901	0.183	0.098	0.026
Quantile tracking	0.903	0.217	0.086	0.000

Table 20: RealToxicityPrompts with GPT-2 continuations, prompt-length order (1 stream; mean and sd over streams). Risk: toxicity probability of the continuation.

Method	Coverage	Max local violation	Mean threshold	Intervention rate
Static	0.940	0.167	0.234	0.000
Rolling	0.905	0.200	0.089	0.000
Adaptive CRC, window W	0.900	0.183	0.122	0.045
Adaptive CRC, matched window	0.900	0.200	0.142	0.067
Kernel weights + update	0.900	0.200	0.111	0.025
Kernel weights, no update	0.906	0.217	0.076	0.000
PromptShift-CRC	0.900	0.183	0.120	0.035
Gate + recency only	0.900	0.183	0.128	0.044
No gate (fixed mix)	0.900	0.183	0.125	0.038
No recency fallback	0.900	0.200	0.111	0.025
No online update	0.906	0.217	0.083	0.000
DtACI	0.906	0.183	0.126	0.023
Density-ratio weights	0.900	0.200	0.117	0.032
Quantile tracking	0.901	0.200	0.082	0.000

Table 21: RealToxicityPrompts with GPT-2 continuations, ascending prompt toxicity (1 stream; mean and sd over streams). Risk: toxicity probability of the continuation.

Method	Coverage	Max local violation	Mean threshold	Intervention rate
Static	0.695	0.600	0.002	0.000
Rolling	0.888	0.233	0.141	0.116
Adaptive CRC, window W	0.899	0.183	0.175	0.155
Adaptive CRC, matched window	0.899	0.183	0.180	0.150
Kernel weights + update	0.899	0.183	0.175	0.152
Kernel weights, no update	0.868	0.283	0.089	0.049
PromptShift-CRC	0.899	0.167	0.175	0.158
Gate + recency only	0.899	0.167	0.176	0.156
No gate (fixed mix)	0.899	0.183	0.172	0.150
No recency fallback	0.899	0.183	0.175	0.152
No online update	0.881	0.250	0.123	0.095
DtACI	0.904	0.200	0.176	0.141
Density-ratio weights	0.899	0.183	0.178	0.154
Quantile tracking	0.888	0.250	0.129	0.087

Table 22: RealToxicityPrompts with GPT-2 continuations, descending prompt toxicity (1 stream; mean and sd over streams). Risk: toxicity probability of the continuation.

Method	Coverage	Max local violation	Mean threshold	Intervention rate
Static	0.997	0.067	0.979	1.000
Rolling	0.913	0.217	0.140	0.117
Adaptive CRC, window W	0.900	0.183	0.120	0.075
Adaptive CRC, matched window	0.900	0.183	0.127	0.085
Kernel weights + update	0.900	0.183	0.117	0.084
Kernel weights, no update	0.935	0.183	0.203	0.187
PromptShift-CRC	0.900	0.183	0.120	0.077
Gate + recency only	0.900	0.183	0.119	0.076
No gate (fixed mix)	0.900	0.183	0.118	0.074
No recency fallback	0.900	0.183	0.117	0.084
No online update	0.923	0.200	0.168	0.140
DtACI	0.907	0.183	0.124	0.085
Density-ratio weights	0.900	0.183	0.119	0.082
Quantile tracking	0.912	0.183	0.163	0.125

Table 23: RealToxicityPrompts with Qwen2.5-0.5B continuations (replication), prompt-toxicity bands in blocks (24 streams; mean and sd over streams). Risk: toxicity probability of the continuation.

Method	Coverage	Post-shift violation	Max local violation	Mean threshold	Intervention rate
Static	0.863 ± 0.093	0.144 ± 0.107	0.369	0.160	0.083
Rolling	0.901 ± 0.005	0.112 ± 0.033	0.284	0.147	0.074
Adaptive CRC, window W	0.900 ± 0.000	0.092 ± 0.011	0.208	0.167	0.116
Adaptive CRC, matched window	0.900 ± 0.000	0.097 ± 0.008	0.209	0.173	0.126
Kernel weights + update	0.900 ± 0.000	0.090 ± 0.012	0.209	0.162	0.111
Kernel weights, no update	0.898 ± 0.014	0.130 ± 0.043	0.342	0.128	0.012
PromptShift-CRC	0.900 ± 0.000	0.092 ± 0.011	0.203	0.168	0.118
Gate + recency only	0.900 ± 0.000	0.092 ± 0.011	0.203	0.168	0.116
No gate (fixed mix)	0.900 ± 0.000	0.092 ± 0.010	0.202	0.167	0.118
No recency fallback	0.900 ± 0.000	0.090 ± 0.012	0.209	0.162	0.111
No online update	0.904 ± 0.009	0.110 ± 0.032	0.281	0.135	0.043
DtACI	0.906 ± 0.001	0.094 ± 0.011	0.220	0.178	0.133
Density-ratio weights	0.900 ± 0.000	0.093 ± 0.010	0.201	0.166	0.115
Quantile tracking	0.900 ± 0.004	0.100 ± 0.018	0.229	0.142	0.023

Table 24: TruthfulQA with Qwen2.5-1.5B-Instruct, category blocks (20 streams; mean and sd over streams). Risk: one minus the probability of the correct answer.

Method	Coverage	Post-shift violation	Max local violation	Mean threshold	Intervention rate
Static	0.883 ± 0.059	0.122 ± 0.063	0.250	1.000	1.000
Rolling	0.895 ± 0.007	0.105 ± 0.011	0.223	1.000	1.000
Adaptive CRC, window W	0.900 ± 0.002	0.099 ± 0.008	0.182	1.000	1.000
Adaptive CRC, matched window	0.900 ± 0.002	0.099 ± 0.006	0.165	1.000	1.000
Kernel weights + update	0.900 ± 0.002	0.101 ± 0.008	0.182	1.000	1.000
Kernel weights, no update	0.901 ± 0.012	0.101 ± 0.013	0.229	1.000	1.000
PromptShift-CRC	0.900 ± 0.002	0.100 ± 0.007	0.168	1.000	1.000
Gate + recency only	0.900 ± 0.002	0.100 ± 0.007	0.169	1.000	1.000
No gate (fixed mix)	0.901 ± 0.002	0.100 ± 0.007	0.167	1.000	1.000
No recency fallback	0.900 ± 0.002	0.101 ± 0.008	0.182	1.000	1.000
No online update	0.906 ± 0.010	0.096 ± 0.012	0.197	1.000	1.000
DtACI	0.915 ± 0.004	0.085 ± 0.008	0.156	1.000	1.000
Density-ratio weights	0.901 ± 0.002	0.100 ± 0.007	0.174	1.000	1.000
Quantile tracking	0.900 ± 0.000	0.100 ± 0.002	0.121	0.998	1.000

Table 25: TruthfulQA with Qwen2.5-1.5B-Instruct, random order, no shift (20 streams; mean and sd over streams). Risk: one minus the probability of the correct answer.

Method	Coverage	Max local violation	Mean threshold	Intervention rate
Static	0.899 ± 0.043	0.186	1.000	1.000
Rolling	0.902 ± 0.004	0.190	1.000	1.000
Adaptive CRC, window W	0.900 ± 0.001	0.170	1.000	1.000
Adaptive CRC, matched window	0.900 ± 0.002	0.165	1.000	1.000
Kernel weights + update	0.900 ± 0.001	0.168	1.000	1.000
Kernel weights, no update	0.906 ± 0.006	0.176	1.000	1.000
PromptShift-CRC	0.900 ± 0.002	0.163	1.000	1.000
Gate + recency only	0.900 ± 0.002	0.164	1.000	1.000
No gate (fixed mix)	0.900 ± 0.002	0.163	1.000	1.000
No recency fallback	0.900 ± 0.001	0.168	1.000	1.000
No online update	0.908 ± 0.004	0.162	1.000	1.000
DtACI	0.915 ± 0.006	0.150	1.000	1.000
Density-ratio weights	0.900 ± 0.001	0.167	1.000	1.000
Quantile tracking	0.900 ± 0.000	0.117	0.999	1.000

Table 26: TruthfulQA with Qwen2.5-1.5B-Instruct, embedding-direction order (forward) (1 stream; mean and sd over streams). Risk: one minus the probability of the correct answer.

Method	Coverage	Max local violation	Mean threshold	Intervention rate
Static	0.897	0.217	1.000	1.000
Rolling	0.905	0.200	1.000	1.000
Adaptive CRC, window W	0.904	0.150	1.000	1.000
Adaptive CRC, matched window	0.904	0.167	1.000	1.000
Kernel weights + update	0.904	0.150	1.000	1.000
Kernel weights, no update	0.904	0.217	1.000	1.000
PromptShift-CRC	0.904	0.167	1.000	1.000
Gate + recency only	0.904	0.150	1.000	1.000
No gate (fixed mix)	0.904	0.167	1.000	1.000
No recency fallback	0.904	0.150	1.000	1.000
No online update	0.908	0.200	1.000	1.000
DtACI	0.916	0.150	1.000	1.000
Density-ratio weights	0.904	0.150	1.000	1.000
Quantile tracking	0.900	0.117	0.999	1.000

Table 27: TruthfulQA with Qwen2.5-1.5B-Instruct, embedding-direction order (reversed) (1 stream; mean and sd over streams). Risk: one minus the probability of the correct answer.

Method	Coverage	Max local violation	Mean threshold	Intervention rate
Static	0.858	0.283	1.000	1.000
Rolling	0.904	0.183	1.000	1.000
Adaptive CRC, window W	0.898	0.167	1.000	1.000
Adaptive CRC, matched window	0.900	0.167	1.000	1.000
Kernel weights + update	0.900	0.167	1.000	1.000
Kernel weights, no update	0.901	0.200	1.000	1.000
PromptShift-CRC	0.900	0.167	1.000	1.000
Gate + recency only	0.900	0.150	1.000	1.000
No gate (fixed mix)	0.900	0.150	1.000	1.000
No recency fallback	0.900	0.167	1.000	1.000
No online update	0.911	0.183	1.000	1.000
DtACI	0.914	0.150	1.000	1.000
Density-ratio weights	0.901	0.150	1.000	1.000
Quantile tracking	0.900	0.117	0.999	1.000

Table 28: TruthfulQA with Qwen2.5-1.5B-Instruct, question-length order (1 stream; mean and sd over streams). Risk: one minus the probability of the correct answer.

Method	Coverage	Max local violation	Mean threshold	Intervention rate
Static	0.876	0.217	1.000	1.000
Rolling	0.894	0.217	1.000	1.000
Adaptive CRC, window W	0.898	0.167	1.000	1.000
Adaptive CRC, matched window	0.900	0.183	1.000	1.000
Kernel weights + update	0.900	0.150	1.000	1.000
Kernel weights, no update	0.900	0.167	1.000	1.000
PromptShift-CRC	0.900	0.150	1.000	1.000
Gate + recency only	0.900	0.150	1.000	1.000
No gate (fixed mix)	0.900	0.150	1.000	1.000
No recency fallback	0.900	0.150	1.000	1.000
No online update	0.905	0.167	1.000	1.000
DtACI	0.915	0.150	1.000	1.000
Density-ratio weights	0.900	0.167	1.000	1.000
Quantile tracking	0.900	0.117	0.999	1.000

Table 29: Frontier gap on TRUTHFULQA streams (post-shift violation minus the Adaptive-CRC window-sweep curve at the same mean threshold; mean over orderings with standard error). Negative is better than Adaptive CRC. Where a method's mean threshold lies outside the range of the window sweep the curve is clipped to its nearest end; the last column repeats the PromptShift-CRC gap over the orderings whose threshold lies inside the range.

Ordering	PromptShift-CRC	Gate + recency only	Kernel + update	DtACI	Density ratio	PromptShift-CRC, inside range
category blocks	-0.0004 (0.0007)	-0.0004 (0.0007)	+0.0003 (0.0004)	-0.0162 (0.0012)	-0.0009 (0.0006)	-0.0004 (0.0007), 17 of 20

Table 30: Proxy streams built from benchmark fields (no model outputs). Coverage with moving-block-bootstrap standard error.

Stream	Method	Coverage	SE	Mean threshold	Max local violation
TruthfulQA	Static	0.994	0.003	0.568	0.033
	Rolling	0.916	0.027	0.401	0.242
	Adaptive CRC, window W	0.909	0.025	0.394	0.217
	Adaptive CRC, matched window	0.904	0.014	0.381	0.175
	PromptShift-CRC	0.907	0.018	0.387	0.175
	Gate + recency only	0.907	0.017	0.385	0.167
	No gate (fixed mix)	0.907	0.018	0.388	0.175
SummEval	Static	0.888	0.007	0.263	0.183
	Rolling	0.900	0.008	0.420	0.158
	Adaptive CRC, window W	0.902	0.006	0.488	0.142
	Adaptive CRC, matched window	0.902	0.006	0.523	0.150
	PromptShift-CRC	0.902	0.006	0.485	0.150
	Gate + recency only	0.902	0.006	0.524	0.150
	No gate (fixed mix)	0.902	0.006	0.487	0.150
LongHalQA	Static	0.205	0.115	0.286	1.000
	Rolling	0.725	0.057	0.890	0.733
	Adaptive CRC, window W	0.897	0.017	0.880	0.258
	Adaptive CRC, matched window	0.897	0.009	0.841	0.175
	PromptShift-CRC	0.897	0.020	0.921	0.258
	Gate + recency only	0.897	0.013	0.854	0.217
	No gate (fixed mix)	0.897	0.019	0.901	0.250
RealToxicityPrompts	Static	0.891	0.005	0.179	0.208
	Rolling	0.896	0.004	0.183	0.175
	Adaptive CRC, window W	0.900	0.003	0.187	0.167
	Adaptive CRC, matched window	0.900	0.002	0.188	0.167
	PromptShift-CRC	0.900	0.003	0.185	0.158
	Gate + recency only	0.900	0.002	0.187	0.167
	No gate (fixed mix)	0.900	0.003	0.186	0.158

Table 31: RealToxicityPrompts across generators, classifiers and target levels (24 band orderings each). PromptShift-CRC minus the matched baseline, post-shift violation with standard error and number of orderings in which PromptShift-CRC is lower, and mean threshold; frontier gaps of PromptShift-CRC and DtACI.

Configuration	PS − matched post-shift	PS lower in	PS − matched threshold	PS lower in	Frontier gap PS	Frontier gap DtACI
GPT-2, toxic-bert, $\alpha = 0.1$ (main)	-0.0052 (0.0011)	19/24	-0.0011	15/24	-0.0010 (0.0010)	-0.0030 (0.0015)
Qwen2.5-0.5B, unbiased-toxic-roberta, $\alpha = 0.1$ (replication)	-0.0050 (0.0015)	16/24	-0.0053	20/24	-0.0003 (0.0007)	-0.0019 (0.0015)
GPT-2, toxic-bert, $\alpha = 0.05$	-0.0039 (0.0009)	19/24	-0.0166	23/24	-0.0010 (0.0009)	-0.0037 (0.0012)
GPT-2, toxic-bert, $\alpha = 0.2$	-0.0031 (0.0019)	13/24	-0.0025	21/24	-0.0017 (0.0015)	+0.0027 (0.0019)

Table 32: MMLU category blocks (24 orderings) at three target levels, all constants as declared. The spread is the range over the four categories of the $(1 - \alpha)$ quantile of the risk. PromptShift-CRC minus the matched baseline in post-shift violation (standard error, number of orderings in which PromptShift-CRC is lower) and in mean threshold; frontier gaps of PromptShift-CRC (over all orderings and over those inside the range of the window sweep), kernel weights with the update, and DtACI.

Level	Spread of target quantile	PS − matched post-shift	PS lower in	PS − matched threshold	Frontier gap PS	PS, inside range	Kernel + update	DtACI
$\alpha = 0.1$ (main)	0.025	-0.0010 (0.0007)	9/24	+0.0010	-0.0007 (0.0006)	+0.0001 (0.0006), 17/24	-0.0007 (0.0006)	-0.0084 (0.0012)
$\alpha = 0.2$	0.077	-0.0016 (0.0009)	12/24	+0.0033	+0.0003 (0.0007)	+0.0007 (0.0007), 20/24	+0.0006 (0.0008)	-0.0031 (0.0013)
$\alpha = 0.3$	0.242	-0.0004 (0.0013)	11/24	-0.0000	+0.0002 (0.0012)	+0.0004 (0.0013), 23/24	+0.0009 (0.0019)	-0.0002 (0.0015)

Table 33: All four L-ARC configurations (the first of each group is the pre-declared primary one). Coverage, frontier gap and the difference from PromptShift-CRC, as in the previous table.

Setting	Configuration	Coverage	Frontier gap	− PS post-shift	− PS threshold
synthetic (mean of the five streams)	L-ARC, authors' defaults ($\eta_1 = 1$, length scale 1)	0.898	+0.0277 (0.0005)	+0.0237 (0.0004)	-0.0069
	L-ARC, $\eta_1 = 1$, length scale matched to h	0.898	+0.0284 (0.0005)	+0.0205 (0.0004)	-0.0179
	L-ARC, $\eta_1 = 0.1$, length scale 1	0.858	+0.1970 (0.0009)	+0.1925 (0.0009)	-0.0074
	L-ARC, $\eta_1 = 0.1$, length scale matched to h	0.860	+0.1936 (0.0009)	+0.1885 (0.0009)	-0.0087
MMLU, category blocks	L-ARC, authors' defaults ($\eta_1 = 1$, length scale 1)	0.900	+0.0001 (0.0010)	+0.0019 (0.0011)	-0.0091
	L-ARC, $\eta_1 = 1$, length scale matched to h	0.900	-0.0004 (0.0010)	+0.0015 (0.0011)	-0.0103
	L-ARC, $\eta_1 = 0.1$, length scale 1	0.900	-0.0007 (0.0012)	-0.0002 (0.0012)	+0.0018
	L-ARC, $\eta_1 = 0.1$, length scale matched to h	0.900	-0.0007 (0.0013)	-0.0002 (0.0013)	+0.0016
MMLU, subject blocks	L-ARC, authors' defaults ($\eta_1 = 1$, length scale 1)	0.900	-0.0002 (0.0004)	+0.0002 (0.0002)	-0.0087
	L-ARC, $\eta_1 = 1$, length scale matched to h	0.900	-0.0003 (0.0004)	+0.0002 (0.0002)	-0.0099
	L-ARC, $\eta_1 = 0.1$, length scale 1	0.900	+0.0002 (0.0008)	+0.0003 (0.0008)	+0.0077
	L-ARC, $\eta_1 = 0.1$, length scale matched to h	0.900	+0.0002 (0.0007)	+0.0003 (0.0007)	+0.0072
RealToxicityPrompts, GPT-2	L-ARC, authors' defaults ($\eta_1 = 1$, length scale 1)	0.900	-0.0033 (0.0033)	+0.0011 (0.0027)	-0.0016
	L-ARC, $\eta_1 = 1$, length scale matched to h	0.900	-0.0036 (0.0032)	+0.0013 (0.0027)	+0.0060
	L-ARC, $\eta_1 = 0.1$, length scale 1	0.909	-0.0006 (0.0094)	+0.0016 (0.0084)	-0.0359
	L-ARC, $\eta_1 = 0.1$, length scale matched to h	0.907	+0.0005 (0.0083)	+0.0026 (0.0073)	-0.0377
RealToxicityPrompts, Qwen2.5-0.5B	L-ARC, authors' defaults ($\eta_1 = 1$, length scale 1)	0.900	+0.0064 (0.0028)	+0.0064 (0.0027)	-0.0107
	L-ARC, $\eta_1 = 1$, length scale matched to h	0.900	+0.0038 (0.0030)	+0.0069 (0.0025)	-0.0050
	L-ARC, $\eta_1 = 0.1$, length scale 1	0.907	+0.0044 (0.0083)	+0.0070 (0.0073)	-0.0431
	L-ARC, $\eta_1 = 0.1$, length scale matched to h	0.905	+0.0048 (0.0076)	+0.0074 (0.0067)	-0.0444
TruthfulQA, category blocks	L-ARC, authors' defaults ($\eta_1 = 1$, length scale 1)	0.901	-0.0008 (0.0011)	-0.0017 (0.0013)	-0.0044
	L-ARC, $\eta_1 = 1$, length scale matched to h	0.901	-0.0007 (0.0009)	-0.0015 (0.0014)	-0.0049
	L-ARC, $\eta_1 = 0.1$, length scale 1	0.901	-0.0000 (0.0013)	-0.0009 (0.0014)	-0.0006
	L-ARC, $\eta_1 = 0.1$, length scale matched to h	0.901	-0.0003 (0.0012)	-0.0012 (0.0015)	-0.0007

Table 34: Frontier gap (post-shift violation minus the Adaptive CRC window-sweep curve at the method's own mean threshold; mean over 24 orderings with standard error) for kernel bandwidths $h/4$ to $2h$, with the gate centre and slope rescaled with the bandwidth. In square brackets, the number of orderings (of 24) whose mean threshold lies inside the range of the window sweep; where it is small the gap is measured against the clipped end of the curve. Top three settings: the declared band-block and category-block orderings. Others: orderings in which blocks of 128 prompts cycle through the regimes.

Setting	Rule	$h/4$	$h/2$	h	$2h$	DtACI (no kernel)
RealToxicityPrompts, GPT-2: band blocks	PromptShift-CRC	+0.0011 (0.0005) [21]	+0.0030 (0.0014) [15]	-0.0010 (0.0010) [19]	-0.0006 (0.0008) [17]	-0.0030 (0.0015) [24]
	Kernel + update	-0.0074 (0.0023) [0]	-0.0024 (0.0014) [8]	-0.0027 (0.0012) [15]	-0.0026 (0.0012) [15]	
RealToxicityPrompts, Qwen2.5-0.5B: band blocks	PromptShift-CRC	+0.0003 (0.0003) [24]	-0.0002 (0.0009) [20]	-0.0003 (0.0007) [20]	-0.0003 (0.0010) [15]	-0.0019 (0.0015) [24]
	Kernel + update	-0.0031 (0.0025) [0]	-0.0002 (0.0012) [4]	-0.0004 (0.0011) [10]	-0.0008 (0.0012) [15]	
MMLU, $\alpha = 0.1$: category blocks	PromptShift-CRC	+0.0011 (0.0009) [22]	-0.0000 (0.0007) [23]	-0.0007 (0.0006) [17]	-0.0006 (0.0007) [14]	-0.0084 (0.0012) [0]
	Kernel + update	+0.0058 (0.0015) [0]	-0.0001 (0.0008) [20]	-0.0007 (0.0006) [14]	-0.0010 (0.0005) [11]	
RealToxicityPrompts, GPT-2: recurring regimes	PromptShift-CRC	-0.0000 (0.0001) [24]	-0.0001 (0.0001) [24]	-0.0001 (0.0000) [24]	+0.0000 (0.0000) [24]	-0.0056 (0.0003) [24]
	Kernel + update	-0.0011 (0.0001) [10]	-0.0001 (0.0000) [5]	+0.0000 (0.0000) [20]	+0.0000 (0.0000) [22]	
RealToxicityPrompts, Qwen2.5-0.5B: recurring regimes	PromptShift-CRC	-0.0001 (0.0000) [24]	-0.0001 (0.0001) [24]	+0.0000 (0.0001) [24]	+0.0001 (0.0000) [24]	-0.0061 (0.0003) [24]
	Kernel + update	-0.0018 (0.0002) [18]	-0.0000 (0.0000) [7]	-0.0000 (0.0000) [20]	+0.0000 (0.0000) [22]	
MMLU, $\alpha = 0.1$: recurring regimes	PromptShift-CRC	-0.0002 (0.0001) [24]	+0.0001 (0.0001) [20]	-0.0002 (0.0001) [16]	-0.0004 (0.0001) [16]	-0.0062 (0.0003) [0]
	Kernel + update	-0.0033 (0.0002) [0]	-0.0006 (0.0001) [21]	-0.0005 (0.0001) [14]	-0.0004 (0.0001) [16]	
MMLU, $\alpha = 0.2$: recurring regimes	PromptShift-CRC	-0.0002 (0.0001) [24]	-0.0002 (0.0001) [24]	-0.0002 (0.0001) [16]	-0.0003 (0.0001) [18]	-0.0043 (0.0004) [2]
	Kernel + update	-0.0033 (0.0001) [0]	-0.0007 (0.0001) [21]	-0.0004 (0.0001) [16]	-0.0004 (0.0001) [15]	
MMLU, $\alpha = 0.3$: recurring regimes	PromptShift-CRC	-0.0001 (0.0001) [24]	-0.0001 (0.0001) [24]	-0.0000 (0.0001) [23]	+0.0001 (0.0001) [20]	-0.0058 (0.0006) [4]
	Kernel + update	-0.0029 (0.0003) [0]	-0.0002 (0.0001) [23]	-0.0002 (0.0001) [17]	-0.0001 (0.0001) [16]	

Table 35: Does the pre-gate drift predict violations? For each stream the steps after the calibration prefix are split by whether the pre-gate drift is in the top decile of that stream. Columns: violation rate in the top decile minus the rate elsewhere, AUC of the drift for the violation indicator (mean and standard error over orderings, or over seeds for the synthetic streams), and the Spearman correlation across shifts between the jump in drift at the boundary and the post-shift violation, with a 95% bootstrap interval. Left: violations of PromptShift-CRC (the primary outcome). Right: violations of the matched baseline (robustness).

Stream	PS: top decile − rest	PS: AUC	PS: jump vs post-shift	Matched: top decile − rest	Matched: AUC	Matched: jump vs post-shift
RealToxicityPrompts, GPT-2 (bands)	-0.044 (0.002)	0.433 (0.002)	-0.63 [-0.77, -0.47]	-0.042 (0.002)	0.431 (0.002)	-0.38 [-0.55, -0.18]
RealToxicityPrompts, Qwen2.5-0.5B (bands)	-0.027 (0.002)	0.437 (0.002)	-0.66 [-0.78, -0.51]	-0.029 (0.002)	0.435 (0.002)	-0.24 [-0.46, -0.00]
MMLU (categories)	+0.024 (0.003)	0.515 (0.002)	-0.25 [-0.45, -0.02]	+0.025 (0.004)	0.517 (0.002)	-0.15 [-0.36, +0.08]
synthetic: pilot	+0.071 (0.005)	0.587 (0.004)	+0.23 [+0.01, +0.42]	+0.074 (0.005)	0.588 (0.004)	+0.37 [+0.16, +0.53]
synthetic: ablation stream	+0.050 (0.004)	0.553 (0.004)	+0.83 [+0.76, +0.87]	+0.054 (0.004)	0.558 (0.004)	+0.83 [+0.74, +0.88]
synthetic: QA topic shift	+0.057 (0.005)	0.565 (0.004)	-0.37 [-0.54, -0.16]	+0.055 (0.004)	0.560 (0.004)	-0.34 [-0.52, -0.14]
synthetic: summarization shift	+0.047 (0.004)	0.562 (0.004)	-0.39 [-0.56, -0.20]	+0.045 (0.004)	0.556 (0.004)	-0.36 [-0.54, -0.17]
synthetic: safety-intent shift	+0.051 (0.005)	0.553 (0.004)	-0.20 [-0.38, +0.00]	+0.050 (0.006)	0.547 (0.004)	-0.15 [-0.36, +0.06]

Table 36: Settings used in every experiment. W is the full window, "recent" the recency window, h the kernel bandwidth, gate centre c and slope κ as in Section 3, γ the step of the online update, and "pool" the length of the history used by the weighted rules. All use $\alpha = 0.1$ except the rows that give another level. The synthetic constants are those of the earlier version of this work and were not changed; the real-stream constants follow rules fixed in advance. "Bound" is the Theorem 2 bound on the distance between coverage and 0.90. For the proxy streams the window lengths follow the rule in the released script (a fraction of the stream length with caps).

Experiment	Prefix	W	Recent	h	Gate c	Slope κ	γ	Pool	Bound
Synthetic pilot	300	300	160	1.8	2.0	2.0	0.025	3W	0.018
Synthetic ablation stream	300	300	160	1.85	2.05	2.2	0.02	3W	0.022
Synthetic prototype tasks (3)	300	280	150	1.85	2.05	2.2	0.02	3W	0.024
MMLU	256	256	128	h (median)	$1.1h$	$2.5/h$	0.02	3W	0.020
RealToxicityPrompts (both generators)	256	256	128	h (median)	$1.1h$	$2.5/h$	0.02	3W	0.012
MMLU, $\alpha = 0.2$ and 0.3	256	256	128	h (median)	$1.1h$	$2.5/h$	0.02	3W	0.018 and 0.016
RealToxicityPrompts (GPT-2), $\alpha = 0.05$ and 0.2	256	256	128	h (median)	$1.1h$	$2.5/h$	0.02	3W	0.013 and 0.011
TruthfulQA	100	100	50	h (median)	$1.1h$	$2.5/h$	0.03	3W	0.043
Proxy: TruthfulQA fields	rule	rule	rule	1.25	1.35	2.5	0.02	3W	0.070
Proxy: SummEval	rule	rule	rule	0.75	0.85	4.0	0.018	3W	0.040
Proxy: LongHalQA	rule	rule	rule	0.55	0.55	5.0	0.018	3W	0.046
Proxy: RealToxicityPrompts fields	rule	rule	rule	0.9	0.95	3.0	0.015	3W	0.017